

Master Thesis

Evaluating Speech-to-IPA Models for Intelligibility Classification of L2 English Pronunciation

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a thesis submitted in fulfillment of the requirements for the degree of

MA Linguistics
(Language and AI)

Vrije Universiteit Amsterdam

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Date	June 25, 2026
Student number	2879447
Word count	19300

Abstract

Providing reliable pronunciation assessment remains a challenge in large-scale language learning environments, where human feedback may be inconsistent and difficult to scale. This thesis investigates whether off-the-shelf speech-to-IPA models can be used for automatic intelligibility classification of English as a second language (L2) pronunciation and whether combining multiple systems can improve classification performance.

Five speech-to-IPA models (IPA-Whisper, MultiIPA, Allosaurus, Wav2Vec2Phoneme-LV60, and Wav2Vec2Phoneme-XLSR) were evaluated using single-word learner recordings from the MOOC *English Pronunciation in a Global World*. The models were first compared against gold-standard IPA transcriptions using both direct transcription matching and feature-based analyses. The resulting phonological information was then used to train ensemble classifiers for intelligibility prediction.

The results showed that individual speech-to-IPA systems provide useful information about pronunciation quality but are not sufficiently accurate as standalone intelligibility classifiers. IPA-Whisper achieved the strongest overall performance among the individual models, while feature-based analyses revealed systematic error patterns, particularly involving aspiration. Many transcription mismatches occurred in productions that were nevertheless judged intelligible by human annotators, highlighting the distinction between pronunciation accuracy and intelligibility. An XGBoost ensemble classifier substantially outperformed all individual systems, achieving F0.5 scores of 0.756 for intelligible pronunciations and 0.672 for unintelligible pronunciations. These results represent a considerable improvement over the best previously reported Whisper-based intelligibility classifier evaluated for this same research project, which achieved corresponding F0.5 scores of 0.600 and 0.588 (Kuan, 2025).

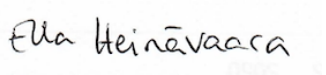
The findings suggest that intelligibility assessment is better understood as a problem of cue integration than direct transcription matching alone. While speech-to-IPA models capture valuable phonological information, their greatest utility appears to lie in serving as components of larger assessment frameworks. The results demonstrate the potential of combining phonological information from multiple systems for automatic intelligibility assessment and contribute to ongoing research on pronunciation evaluation in language learning contexts.

Declaration of Authorship

I, Ella Heinävaara, declare that this thesis, titled *Evaluating Speech-to-IPA Models for Intelligibility Classification of L2 English Pronunciation* and the work presented in it are my own. I confirm that:

- This work was done wholly or mainly while in candidature for a Master degree at this University.
- Where any part of this thesis has previously been submitted for a degree or any other qualification at this University or any other institution, this has been clearly stated.
- Where I have consulted the published work of others, this is always clearly attributed.
- Where I have quoted from the work of others, the source is always given. With the exception of such quotations, this thesis is entirely my own work.
- I have acknowledged all main sources of help.
- Where the thesis is based on work done by myself jointly with others, I have made clear exactly what was done by others and what I have contributed myself.

Date: June 25, 2026

Signed: 

Acknowledgments

First and foremost, I would like to thank my parents for their unwavering support throughout my studies. Their encouragement, patience, and belief in me have made this thesis, and indeed my entire academic journey, possible. I am deeply grateful for everything they have done for me.

I would also like to thank my friends for their support, encouragement, and for providing much-needed distractions throughout the writing process. Their companionship and countless conversations made the more challenging moments considerably easier and helped keep me sane during the final stages of this thesis. I would like to especially thank Maria for giving me feedback and being there for me.

I would like to thank my supervisor, Luis, for overseeing this thesis.

Finally, I would like to thank everyone involved in the *English Pronunciation in a Global World* project and the internship that inspired this work. I would like to especially thank the creator of the course, Laura Rupp. Their efforts in creating and annotating the dataset made this research possible.

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Chapter 1

Introduction

1.1 Problem Definition

Providing meaningful feedback remains a persistent challenge in language learning, particularly in large-scale online environments. This thesis is part of an internship with the Centre of Global English at the Vrije Universiteit Amsterdam. The Centre for Global English hosts a MOOC, *English Pronounciation in a Global World*[1], in which the learners are asked to evaluate each other's speech through peer review systems. While this approach enables feedback at a large scale, it also introduces variability and inconsistency, as learners may lack the expertise required to assess pronunciation accurately. As a result, feedback can be limited in both reliability and phonetic detail.

One aspect of pronunciation that is particularly important to assess is intelligibility, that is, whether a listener can successfully understand the speaker's intended message. Previous work within the same internship project explored the use of Automatic Speech Recognition (ASR) systems for intelligibility classification, demonstrating that automatically generated transcriptions can provide useful information about pronunciation quality. However, conventional ASR systems are primarily designed to recover words rather than represent how speech is pronounced. As a result, they may overlook pronunciation differences that are relevant for intelligibility assessment.

Recent developments in speech processing have led to the emergence of models that directly transcribe speech into International Phonetic Alphabet (IPA) representations. Unlike conventional ASR systems, these speech-to-IPA models aim to capture pronunciation at the segmental level, making them potentially useful for pronunciation assessment and phonetic analysis. At the same time, little is known about how effectively such models can be used for intelligibility classification, particularly in the context of non-native English speech.

To address this question, the present study evaluates four off-the-shelf speech-to-IPA models: wav2vec2phoneme, MultIPA, Allosaurus, and IPA-Whisper, on recordings of non-native English single-word pronunciations. Model outputs are evaluated against gold-standard multivalued IPA annotations (i.e., each word may have multiple acceptable IPA transcriptions). The models are assessed both in terms of their ability to represent pronunciation accurately and their usefulness for intelligibility classification.

The evaluation proceeds in three stages: first, the models are compared using direct IPA string comparison against gold annotations, providing a baseline measure of

[1] https://www.futurelearn.com/courses/english-pronunciation

phonetic accuracy. Second, the analysis is extended to a feature-based level, where IPA outputs are decomposed into phonetic features. This allows for a more detailed assessment of model performance, revealing which aspects of pronunciation (e.g, voicing, aspiration, vowel quality) are captured or missed. Third, based on the observed strengths and weaknesses of each model, an ensemble approach is proposed that combines complementary information across models. The performance of this ensemble model is then evaluated on a test set. For a proper comparison, the performance is compared to the best-performing Speech-to-IPA model, which is also evaluated on the test set.

1.2 Research Question

The goal of this research was to investigate whether off-the-self speech-to-IPA models be used for intelligibility classification.

This leads to two main research questions:

1. How effectively can different off-the-shelf speech-to-IPA models be used for intelligibility classification?

This question is addressed through both surface-level IPA comparison and feature-based analysis to investigate the models' strengths and weaknesses.

2. To what extent can complementary strengths of different speech-to-IPA models be combined in an ensemble model to improve intelligibility classification?

1.3 Overview of the Paper

The rest of the thesis is organized as follows: Chapter 2 reviews the relevant literature on phonetics, intelligibility in English as a second language (L2), and speech-to-IPA models. The chapter also discusses previous research conducted within the wider study project, introduces the phonological feature framework used in the study, and presents the study hypotheses. Chapter 3 describes the methodology, including the dataset, preprocessing procedures, speech-to-IPA models, evaluation methods, feature-based analysis, and ensemble model construction.

The next three chapters are all results on the development set. Chapter 4 presents the baseline comparison of the speech-to-IPA models on the development set using direct IPA matching. Chapter 5 extends this analysis to the feature level by examining model-specific error patterns, common phoneme substitutions, and the contribution of individual phonological features to model performance. Chapter 6 presents the performances of the ensemble models.

Chapter 7 reports the final evaluation on the test set, comparing the best-performing individual speech-to-IPA model with the best-performing ensemble model. Chapter 8 provides an error analysis of both systems, focusing on confusion patterns, word-level performance, and phonological properties associated with classification errors. Chapter 9 discusses the findings in relation to previous research, addresses the research questions, considers limitations of the study, and outlines directions for future work. Finally, Chapter 10 concludes the thesis.

Chapter 2

Literature Review

2.1 Phonetics and Phonemes

Speech is composed of smaller linguistic units that together form meaningful language. At the most abstract level, spoken words are built from phonemes, which are commonly defined as the smallest units capable of distinguishing meaning between words. For example, the difference between /p/ and /b/ distinguishes the English words *pat* and *bat*. Phonemes are therefore often described as the "building blocks" of language, a phrase coined by Ferdinand de Saussure (1959).

Importantly, phonemes are language-specific (Ladefoged and Johnson, 2014). Each language organizes speech sounds into its own phonological system, meaning that distinctions that are meaningful in one language may not exist in another. English is commonly described as having around 44 phonemes, although the exact number varies depending on dialect and analytical approach (Roach, 2009). Consequently, second-language learners must acquire not only new vocabulary and grammar but also new phonological contrasts and articulatory patterns.

Phonetics is the field concerned with the physical properties of speech sounds, including how they are produced, transmitted, and perceived. Articulatory phonetics specifically examines how speech sounds are created through movements of the vocal tract. Consonants are commonly categorized according to features such as place of articulation (where constriction occurs in the vocal tract), manner of articulation (how airflow is modified), and voicing (whether the vocal folds vibrate). Vowels are typically described through tongue height, tongue backness, lip rounding, and vowel length (Ladefoged and Johnson, 2014). These articulatory dimensions form the basis of many phonological feature systems and are central to computational representations of speech sounds.

While learners may produce non-native pronunciations, not all deviations from target-like pronunciation lead to communication difficulties. In many cases, speech may remain fully understandable despite a noticeable foreign accent. This distinction motivates the concept of intelligibility, which has become a central focus in second language pronunciation research.

2.2 Intelligibility in L2 English

Intelligibility is a central concept in second language research (Munro and Derwing, 1999), but it is also one of the most debated (Nelson et al., 2011). Broadly, intelligibility refers to the extent to which a listener can correctly understand a speaker's intended message. However, it is important to distinguish intelligibility from related constructs such as accent and comprehensibility, which capture different aspects of speech perception.

Munro and Derwing (1999) provides one of the most widely accepted frameworks by distinguishing key dimensions: foreign accent, comprehensibility, and intelligibility. A foreign accent is defined as how different speech sounds are from a native norm; comprehensibility is defined as how easy speech is to understand, and intelligibility is how much speech is actually understood. This distinction is crucial, as speech can be strongly accented, yet still fully intelligible. In other words, deviation from native pronunciation does not necessarily result in a communication breakdown.

In the context of English, a global language spoken by many, this distinction has led to a shift away from native-like pronunciation as the primary goal. Instead, researchers increasingly emphasize intelligibility as the key criterion for successful communication. Levis (2005) contrasted two approaches to pronunciation teaching: the Nativeness Principle and the Intelligibility Principle. Levis argues that aiming for native-like pronunciation is both unrealistic and unnecessary. Instead, pronunciation should be evaluated on the basis of whether it supports successful communication. Importantly, he reinforces that accent is not a reliable predictor of intelligibility and that not all pronunciation features contribute equally to understanding.

Similarly, earlier, Jenkins (2000, 2002) demonstrated that only a subset of phonological features consistently affects intelligibility, particularly in interactions between non-native speakers. Specifically, her findings show that communication breakdowns are often caused by failures in key segmental contrasts and syllable structure, rather than by overall accent or fine phonetic detail. This leads to the proposal of the Lingua Franca Core (LFC), which identifies the essential features to maintain intelligibility.

Table 2.1 below adapts the LFC specifically for single-word annotation. It combines findings of Jenkins (2000) and Jenkins (2002).

Table 2.1: Lingua Franca Core for single-word utterances

Feature	**Example**
Consonant contrast	*pat* vs. *bat*
(Initial) Consonant clusters[1]	*play* vs. *pay*
Vowel length contrasts	*ship* vs. *sheep*
Distinct phoneme realization	/v/ vs. /b/
Pre-fortis clipping	bet vs. bed
Rhoticity over non-rhotic accents	/bəːd/ > /bɔːd/
Word-initial /p, t, k/ are aspirated	/p^{h}ot/

[1]This might seem similar to the first feature, *consonant contrast*. The difference is that *consonant contrast* is on a phonological level, meaning that a change in the phoneme changes the word meaning, while *distinct phoneme realization* is on a phonetic level, and concerns how clearly these sounds are articulated, namely, are they pronounced clearly enough to be distinguished.

The emphasis on intelligibility rather than native-like pronunciation has important implications for automatic speech analysis. If only certain phonetic features consistently affect intelligibility, then automatic systems may benefit from focusing on these features rather than attempting to model all phonetic detail equally.

2.3 Speech Processing and Phoneme Recognition

Speech processing encompasses computational methods for analyzing, recognizing, synthesizing, and generating spoken language (Jurafsky and Martin, 2026). These systems transform acoustic speech signals into linguistic representations and have become increasingly important in both speech technology and linguistic research.

One task in speech processing is phoneme recognition, also known as Automatic Phoneme Recognition (APR), which focuses on identifying phonemes or phonetic segments directly from speech signals (Malakar and Keskar, 2021). Because phonemes encode contrastive sound categories, phoneme-level representations provide a useful bridge between raw acoustic information and higher-level linguistic analysis.

Previous research has approached speech-to-phoneme conversion from several related perspectives, including phoneme recognition, pronunciation assessment, and phonetic alignment. Malakar and Keskar (2021) provides a broad overview of machine-learning approaches to phoneme recognition, describing how systems map acoustic speech signals onto phonemic categories using both traditional statistical methods and modern neural architectures. Unlike conventional ASR systems, which are primarily designed to recover words, phoneme recognition systems focus on segmental representations and are therefore particularly relevant for pronunciation-oriented research.

Research has explored the role of APR in broader linguistic analysis. Martin et al. (2024) investigated whether automatic phoneme recognition systems are suitable for analyzing spontaneous French speech, emphasizing the importance of phoneme recognition accuracy and segmentation quality for linguistic research. Similarly, recent work has focused on multilingual and low-resource phoneme recognition. Siminyu et al. (2021) fine-tuned the multilingual phonetic recognition system Allosaurus on Bukusu and Saamia, two Luhya language varieties spoken in Kenya, demonstrating how multilingual pretrained models can be adapted to generate phonetic transcriptions for under-resourced languages.

2.4 Previous Research within the Wider Project

The present study builds on previous internship research conducted at the Centre for Global English (CGE), which explored the use of Automatic Speech Recognition (ASR) systems for pronunciation intelligibility assessment.

First, Zou (2024) investigated whether OpenAI's Whisper `base.en` model could be used for binary intelligibility classification of learner speech. The study compared Whisper transcriptions with the target words to classify pronunciations as either intelligible or unintelligible. Zou also introduced a phonetics-based classification method using CMUDict to handle homophones better, such as *meet* and *meat*. The results showed that phonetic-based scoring outperformed simple text matching, suggesting that intelligibility is better captured by phonetic similarity than by exact spelling

matches. However, the study also found that Whisper struggled with isolated-word recognition and often produced overgenerated or nonsensical transcriptions.

Building on this work, Kuan (2025) investigated whether continual pretraining on isolated single-word utterances could improve Whisper's performance for intelligibility classification. Unlike Zou's study, which used the original Whisper model, Kuan continually pretrained several Whisper models using isolated English word recordings from Mozilla Common Voice and Wiktionary. The study evaluated both transcription accuracy and intelligibility classification using exact-match and phonetic similarity scoring methods. The results showed that continual pretraining improved isolated word recognition and also improved intelligibility classification performance. The study also highlighted the importance of phonetic features such as aspiration, vowel substitutions, and word-final consonant reduction in assessing intelligibility.

Together, these studies demonstrate how ASR systems can be adapted for research on pronunciation intelligibility and provide the methodological foundation for the present study.

These earlier studies motivate the present research in two important ways. First, they demonstrate that pronunciation intelligibility cannot be adequately modeled through orthographic comparison alone. Two pronunciations may differ in spelling while remaining phonetically very similar, whereas small phonetic differences may strongly affect intelligibility. Second, they show that phonetic information provides a more linguistically meaningful representation for pronunciation assessment than raw text transcriptions alone.

For this reason, the current study moves from orthographic ASR outputs toward direct phonetic representations. Rather than comparing written transcriptions, the present study compares automatically generated IPA transcriptions against gold-standard IPA annotations. This allows pronunciation differences to be evaluated at the segmental and articulatory levels rather than through spelling-based similarity. The primary focus of the study is therefore the comparison between predicted and gold IPA sequences and the extent to which phonetic similarity can capture intelligibility-related variation in learner speech.

To generate these phonetic representations, the study employs several speech-to-IPA and phoneme recognition models. These models automatically convert acoustic speech input into phonetic transcriptions represented in IPA. Unlike conventional ASR systems, which are optimized for recovering words or sentences, speech-to-IPA models focus directly on phonetic realization and pronunciation detail. This makes them particularly suitable for pronunciation analysis, where the goal is not necessarily to recover the intended word, but rather to analyze how it was pronounced.

2.5 Models Explored in this Study

This study explores four types of speech-to-IPA systems: MultIPA, Wav2Vec2Phoneme, Whisper IPA, and Allosaurus, each representing a different approach to phonetic modeling. Two different types of Wav2Vec2Phoneme on models were used, one trained on English data and one on multilingual data. This makes it five models in total.

The models examined in this study operate at different levels of speech representation. Phonetic representations aim to capture the surface realization of speech sounds. For example, a word such as *pit* may be transcribed as [p^hɪt], preserving details

such as aspiration. Phonological representations, in contrast, encode language-specific categories rather than fine phonetic detail. For example, Wav2Vec2Phoneme outputs eSpeak phonemes, which are subsequently converted to IPA for comparison, resulting in a representation closer to underlying phonological units than to narrow phonetic transcription. Some systems combine characteristics of both levels and are therefore described as hybrid. Table 2.2 summarizes the speech-to-IPA models considered in this study, their underlying architectures, output formats, and representation levels.

Table 2.2: Speech-to-IPA models explored in this study.

Model	**Base Model**	**Output Example**	**Representation**
Wav2Vec2Phoneme	Wav2Vec 2.0	eSpeak (/pIt/) → IPA [pɪt]	Phonological
MultIPA	Wav2Vec 2.0	[pʰɪt]	Phonetic
Allosaurus	Acoustic model	[pʰɪt]	Phonetic
IPA Whisper	Whisper	[pʰɪt]	Hybrid

Each individual model is now presented.

2.5.1 Wav2Vec2Phoneme

The Wav2Vec2Phoneme model by Xu et al. (2022) builds on the self-supervised speech model wav2vec 2.0 (Baevski et al., 2020), which learns speech representations directly from raw audio. Wav2Vec 2.0 uses a convolutional neural network (CNN) to encode the audio signal and a Transformer to model contextual information across time. After pretraining on large amounts of unlabeled speech, the model can be fine-tuned for downstream tasks such as speech recognition.

Wav2Vec2Phoneme adapts this architecture for phonetic transcription by fine-tuning the pretrained model to predict phoneme sequences rather than orthographic text. The model is trained using Connectionist Temporal Classification (CTC) loss, which allows it to align audio with phoneme labels without requiring precise segmentation.

Importantly, the output of Wav2Vec2Phoneme consists of phonemes, which are language-specific units representing abstract sound changes. As a result, the model operates at a phonological level of representation, and its output depends on the phoneme inventory used during training. It outputs pronunciations using the eSpeak phoneme set[2], a language-dependent symbolic representation originally developed for speech synthesis. These outputs were subsequently converted to IPA to facilitate comparison with the other models and the gold-standard transcriptions. Because eSpeak employs its own phoneme inventory, each eSpeak symbol was mapped to its IPA equivalent using a deterministic one-to-one conversion (e.g., `pIt` → /pɪt/).

Wav2Vec2Phoneme is included as a phoneme-based baseline for comparison with IPA models and to highlight the difference between phonological and phonetic representations.

Two versions of this model will be compared. The first one, LV-60[3], is trained on approximately 60,000 hours of primarily English speech data. The other one, XLSR-53[4], is multilingual and trained with speech data from 53 different languages. Both models were included to see whether the model-training language matters.

[2]https://espeak.sourceforge.net/index.html
[3]https://huggingface.co/facebook/wav2vec2-lv-60-espeak-cv-ft
[4]https://huggingface.co/facebook/wav2vec2-xlsr-53-espeak-cv-ft

2.5.2 MultIPA

MultIPA[5] by Taguchi et al. (2023) is also based on a pretrained wav2vec 2.0 model. Like Wav2Vec2Phoneme, it uses a wav2vec-based system. MultIPA uses a Transformer architecture and is trained with CTC loss to align audio with output sequences. Unlike Wav2Vec2Phoneme, MultIPA is trained to produce IPA. This means that the model operates at a phonetic level, aiming to represent actual speech sounds rather than abstract categories. Because IPA provides a standardized, language-independent representation, the output of MultiIPA is directly comparable across languages.

2.5.3 Allosaurus

The Allosaurus[6] model by Li et al. (2020) is a multilingual system designed for universal phonetic transcription. It is based on a supervised acoustic model rather than a pretrained self-supervised framework, as in the previous models. The model is trained directly on speech and phoneme transcripts, using a shared encoder to learn acoustic representations across multiple languages.

A key feature of Allosaurus is its explicit distinction between phones and phonemes. The model first predicts a distribution over universal phone inventory using a shared acoustic encoder. These phone representations are then mapped to phonemes through an allophone layer, which captures how sounds are realized differently across languages. This allows the model to incorporate linguistic knowledge about sound variation directly into its architecture.

2.5.4 IPA Whisper

IPA-Whisper[7] is based on Whisper (Radford et al., 2022), which is a large-scale speech recognition model trained on hundreds of thousands of hours of audio-text data. It uses an encoder-decoder Transformer architecture, in which the encoder processes audio as log-Mel spectrograms and the decoder generates text. A key strength of Whisper is its robustness and ability to generate across languages and domains without additional fine-tuning.

Although Whisper is originally designed to output orthographic text, its architecture is flexible and can be adapted to other output representations, such as IPA, as shown here with IPA-Whisper. This output difference was achieved by modifying the output vocabulary and training the model on speech data paired with phonetic transcriptions. The IPA-Whisper model is fine-tuned on synthetic IPA data.

2.6 Lexical Influence in Speech-to-IPA models

Although speech-to-IPA models aim to produce phonetic representations, they are not entirely free from lexical influence. This is because they are typically trained on speech derived from real words, meaning they implicitly learn phonotactic constraints and language-specific sound distributions. As a result, model output may reflect not only acoustic input but also learned expectations about valid sound sequences. Evidence

[5] https://github.com/ctaguchi/multipa
[6] https://github.com/xinjli/allosaurus
[7] https://huggingface.co/neurlang/ipa-whisper-base

for this can be found in L2 ASR research, where models perform poorly when speech deviates from expected patterns (Emara and Shaker, 2024; Sarymsakova and Rodilla, 2025). This suggests that even phonetic models may encode a degree of lexical or phonological bias. While lexicality bias is an important consideration, it does not necessarily undermine the usefulness of speech-to-IPA models for the present study. It is rather quite the opposite. It highlights the need to interpret model outputs within a broader linguistic framework, particularly regarding intelligibility.

2.7 Feature representation

As hinted at earlier, this study will also apply feature-based analysis to provide a more detailed assessment of the models' strengths and weaknesses. The feature representation used in this study follows the general logic of articulatory feature systems such as PanPhon, which encode IPA segments as bundles of phonological features rather than as isolated symbols. PanPhon maps more than 5000 IPA segments to multidimensional articulatory feature vectors Mortensen et al. (2016). Unlike one-hot symbol encodings, feature-based representations encode shared articulatory properties between sounds, allowing phonetic similarity to be represented directly. However, the full PanPhon feature inventory contains over 20 features, including highly fine-grained distinctions such as [± distributed], [±strident], and [±constricted glottis]. While these distinctions are valuable for multilingual phonological modeling, they are unnecessarily detailed for the goals of the present study. Instead, the current feature set adopts a reduced articulatory representation, focusing only on dimensions directly relevant to broad phonetic categorization and cross-segment similarity. This simplification improves interpretability and reduces sparsity while still preserving the major articulatory contrasts necessary for analysis.

The features are split into vocalic and consonantal features and are introduced in the following chapter, Chapter 3.

2.8 Output Representations and Comparability

A central challenge in comparing Speech-to-IPA models lies in the heterogeneity of their output representations. Different models produce outputs at different levels of abstraction, and some even use different symbol sets (e.g., eSpeak for Wav2Vec2Phoneme vs. different IPA levels for the rest). These differences are not incidental but reflect underlying model architectures and training objectives. To enable meaningful comparison, outputs must therefore be transformed into a shared representation, namely IPA here.

In addition to differences in output format and representational level, the models also vary in the extent to which they encode fine phonetic detail. Models such as Allosaurus and MultiIPA can produce relatively narrow phonetic transcriptions, including features like aspiration and subtle vowel variation. At the same time, Wav2Vec2Phoneme operates at the phoneme level and cannot represent such allophonic detail. IPA-Whisper occupies an intermediate position, depending on how strongly phonetic detail is reflected in its training data. These differences mean that model outputs vary not only in correctness, but also in the degree of phonetic specificity.

Following Jenkins (2000, 2002), certain phonetic features, such as the aspiration of word-initial voiceless stops, are considered relevant for intelligibility in English. In the present study, such features are therefore treated as part of the target representation and included in the gold-standard annotations. Consequently, models that fail to encode these features are penalized, even if they correctly capture the underlying phonological form. This results in a stricter evaluation that distinguishes between phonological accuracy (correct segmental structure) and phonetic completeness (accurate realization of surface form).

At the same time, these differences suggest that models may offer complementary strengths. Phoneme-based systems may produce stable and consistent segmental representations, while phone-level or IPA-based models may better capture fine phonetic detail. For this reason, the present study not only compares individual model performance but also considers whether combining outputs from multiple models can yield improved representations. This allows the analysis to assess both the relative strengths of individual systems and the extent to which different modeling approaches contribute distinct and potentially complementary information.

2.9 Ensemble Methods for Pronunciation Assessment

Previous sections have shown that speech-to-IPA systems differ substantially in their representations. Some models prioritize stable phonemic structure, while others capture finer phonetic detail. At the same time, previous research has shown that ASR systems vary substantially in their performance on non-native speech and are sensitive to different phonological characteristics of accented pronunciation (Emara and Shaker, 2024). Differences in model architecture and training data may therefore lead systems to capture different aspects of non-native pronunciation.

This raises the possibility that different speech-to-IPA systems may provide partially complementary information for pronunciation assessment. Phoneme-based systems may provide stable segmental structure, while phone-level systems may better capture fine phonetic detail relevant for intelligibility. Because the models explored in the present study operate at different representational levels and are trained on different types of speech data, combining their outputs may enable pronunciation judgments to draw on multiple sources of phonological information simultaneously.

Ensemble learning provides one approach for combining such complementary model behavior. Rather than relying on the predictions of a single recognizer, ensemble methods combine the outputs of multiple systems to improve robustness and predictive performance. Ensemble approaches are widely used in machine learning because individual models often capture different aspects of the data and therefore make partially different errors.

Several ensemble approaches were explored in the present study, including logistic regression, random forest classifiers, and gradient-boosted decision trees. More on this will be discussed later in Section 3.9 of Chapter 3.

2.10 Hypotheses

This study addresses two research questions as stated earlier. Here they are reiterated.

The first research question asks how well different speech-to-IPA models represent non-native English single-word pronunciations compared to gold-standard annotations. This question is addressed through both direct IPA comparison and feature-based evaluation.

The second research question asks whether the complementary strengths of different speech-to-IPA models can be combined in an ensemble model to improve phonetic representation. This question is addressed by using the development set to identify the model-specific strengths and weaknesses, and then evaluating the resulting ensemble model on a test set.

Based on the literature and the model's characteristics, the following hypotheses are proposed for the first research question.

First, models that operate at a phonetic level are expected to outperform phoneme-based models. Phonetic models aim to capture fine-grained acoustic detail, whereas phoneme-based models represent more abstract, language-specific categories. As a result, phoneme-based systems are limited in their ability to encode important variation such as aspiration and vowel quality. Therefore, it is expected that MultIPA, Allosaurus, and IPA-Whisper will outperform Wav2Vec2Phoneme in both surface-level IPA comparison and feature-based evaluation.

Second, the type of training data is expected to influence model performance. Models trained on multilingual and phonetically diverse data are likely to be better equipped to handle non-native speech, as they are exposed to a wider range of phonetic realizations. In contrast, models trained on synthetic or phoneme-level targets may tend to normalize pronunciations toward more canonical forms. Therefore, it is expected that models trained on real multilingual phonetic data, such as MultIPA and Allosaurus, will better capture non-native variation than IPA-Whisper, which relies on synthetic IPA training data.

Third, a trade-off between stability and phonetic detail is expected. Phoneme-based models are likely to produce more consistent and stable segmental outputs due to their reliance on discrete phonological categories. However, this consistency comes at the cost of reduced sensitivity to fine phonetic variation. In contrast, phonetic models may capture more detail but exhibit greater variability in their outputs.

For the second research question, it is hypothesized that an ensemble model combining outputs from multiple speech-to-IPA systems will outperform individual models in pronunciation assessment accuracy. Since the evaluated systems differ in representational level, phonetic specificity, and error patterns, combining their outputs is expected to provide complementary information that improves robustness and the prediction of intelligibility.

Chapter 3

Methodology

3.1 Pipeline

Figure 3.1 presents the overall methodological pipeline used in the study, which consisted of three main stages.

First, the Speech-to-IPA transcription models were evaluated using direct IPA comparison against manually created gold-standard pronunciations. This stage included data preprocessing (Section 3.3), Speech-to-IPA model inference (Section 3.4), IPA normalization and postprocessing (Section 3.5), and IPA-based evaluation on the development set (Sections 3.6 and 3.7). These stages are shown in blue in Figure 3.1

Second, the model predictions were analyzed using simple phonological feature representations (Section 3.8). This section is in green in the figure.

Finally, the outputs from the individual models and the feature-based error representations were combined into ensemble classifiers (Section 3.9). Multiple ensemble approaches were evaluated on the development set, after which the best-performing ensemble model was trained on the full development set (henceforth DEV) and evaluated on the test set (henceforth TEST). This section is in purple in the figure.

All code used for preprocessing, model evaluation, and ensemble training was implemented in Python.[1]

3.2 Data Description

The data used in this project originates from audio recordings by students enrolled in *the MOOC English Pronunciation in a Global World*[2]. As part of the course, the students recorded themselves reading a list of 52 words[3] that were submitted in MP3 format. The list of words read by the students was adapted from Hughes et al. (2012) and is shown in Table 3.1. The recordings were split into single-word audios. The dataset consists of approximately 10,000 items annotated with a binary label indicating whether they are intelligible or unintelligible. The binary annotation was performed by both the MOOC course instructor and research assistants, and it follows the previously introduced LFC by Jenkins (2000, 2002) (see Table 2.1).

[1] The code used in this thesis is available at: `https://github.com/ellaheii/MA-Thesis`

[2] https://www.futurelearn.com/courses/english-pronunciation

[3] Some iterations have 53 words, with also having the word "*wait*" in the word list. Still, since it is a homophone for "*weight*", it was treated as equivalent to "*weight*" in the analysis and mapped to the same target form to ensure consistency in comparison with the gold standard.

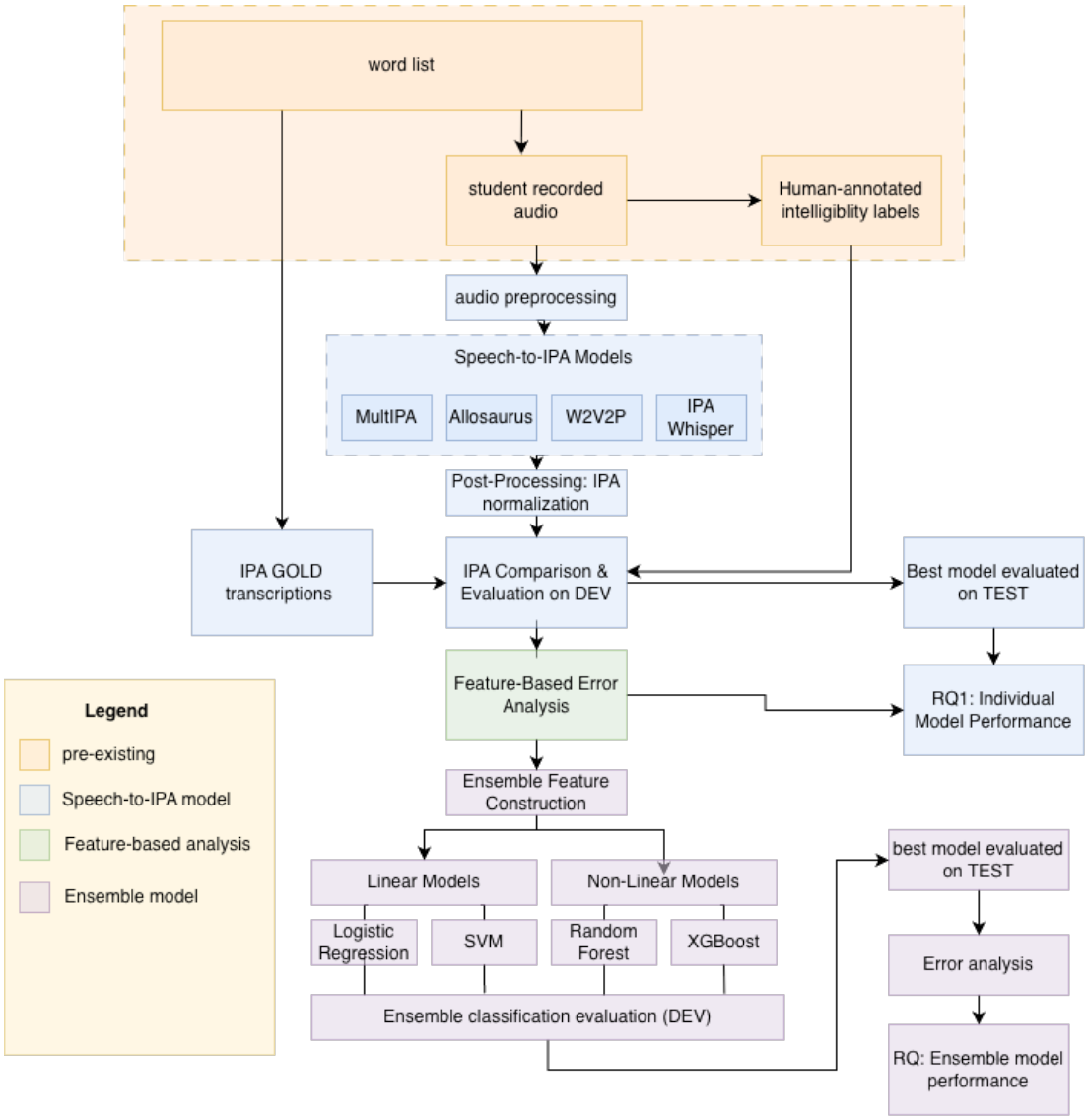


Figure 3.1: Pipeline of the study

Table 3.1: The List of Words read by students

1. pit	14. beer	27. pull	40. plate
2. pet	15. bear	28. pool	41. weight
3. pat	16. bird	29. pole	42. poor
4. put	17. bard	30. Paul	43. pour
5. putt	18. board	31. doll	44. pore
6. pot	19. city	32. cot	45. paw
7. bee	20. seedy	33. caught	46. tide
8. bay	21. hat	34. fir	47. tied
9. buy	22. dance	35. fern	48. pause
10. boy	23. daft	36. fur	49. paws
11. boot	24. half	37. fair	50. meet
12. boat	25. father	38. nose	51. meat
13. bout	26. farther	39. knows	52. mate

The data was split into a development and a test set. The development set had 2001 items (19.5%), and the test set had 8216 items (80.5%). The sets were created so that there would be an even distribution of intelligible and unintelligible pronunciations across the datasets. For both sets, the intelligible class was slightly bigger with 56.3%, and 43.7% for the unintelligible class. The word-level label distributions for both sets can be seen in Figure A1 in Appendix A. The Figure shows the proportion of unintelligible recordings for each target word. As the labels are binary, the proportion of intelligible recordings is complementary and is not shown separately.

The data statement can be seen in Appendix D.

3.3 Preprocessing

Firstly, each word on the word list was transcribed to IPA following Jenkins (2000, 2002). Since more than one pronunciation can be acceptable for some words, multiple gold-standard variants were included where relevant. This list can be found in Appendix B.

The audio files were converted to .wav files for compatibility with the models. Recordings were resampled to 16kHz and converted to mono, as this format is required or preferred by the models used. In addition, Allosaurus required audio input to be provided as a file path rather than a file; thus, to accommodate this, audio was temporarily written to disk.

The filename structure was used to automatically extract both the target word and the human intelligibility label for each item. Filenames followed the pattern `seriesXX-sXXXXX-word-label.mp3`, where the first part identifies the recording series and speaker, the third field contains the target word, and the final field encodes the human intelligibility label. For example, in the filename `series03-s01234-bear-1.mp3`, the target word *bear* and the label `1` (intelligible) were extracted directly from the filename. This information was subsequently used to associate each recording with its corresponding lexical item and binary intelligibility judgment.

3.4 Models

Five model outputs were evaluated in this study. They are the following:

- **IPA-Whisper**
  - Hugging Face: `neurlang/ipa-whisper-medium`
- **MultIPA**
  - Hugging Face: `ctaguchi/multipa`
- **Wav2Vec2Phoneme**
  - Hugging Face: `facebook/wav2vec2-lv-60-espeak-cv-ft`
  - Hugging Face: `facebook/wav2vec2-xlsr-53-espeak-cv-ft`
- **Allosaurus**
  - GitHub: `xinjli/allosaurus`

3.5 Postprocessing

After model inference, all outputs were normalized for comparability. Stress and tone markers, as well as non-essential diacritics, were removed. Non-essential diacritics included dentalization (◌̪) or lowering of a vowel (◌̞). Relevant phonetic information, such as aspiration and vowel length, was kept. Character-level differences between models were also standardized, for example, converting ASCII to IPA (e.g., "g" → "g").

Wav2Vec2Phoneme models output eSpeak-style symbols rather than IPA directly; thus, these outputs were converted into IPA before evaluation.

Rhotic vowels were represented consistently across all outputs by decomposing them into a vowel followed by a rhotic consonant. This made rhoticity easier to compare across models.

So they were decomposed following this form:

$$\text{rhotic vowel} \rightarrow \text{vowel} + \text{rhotic consonant}$$

Here are a few examples:

(a) /ɚ/ → /ə/ & /ɹ/

(b) /ɝ/ → /ɜ/ & /ɹ/

Finally, the transcriptions were tokenized into phoneme-level units. Multi-character segments, such as affricates, long vowels, aspirated consonants, and diphthongs, were treated as single units.

The same postprocessing steps were applied to the GOLD IPA transcriptions.

3.6 Evaluation Metrics

Model performance was evaluated using precision, recall, F1 score, and F0.5 score. These metrics were used consistently throughout the direct IPA evaluation, the feature-based analysis, and the ensemble model evaluation. All metrics were calculated separately for the intelligible and unintelligible classes.

Following previous internship research (Zou, 2024; Kuan, 2025), the unintelligible class was treated as the positive class. Thus, instances where both the GOLD label and the prediction were unintelligible were counted as **True Positives (TP)**. Instances where the GOLD label was unintelligible but the prediction was intelligible were counted as **False Negatives (FN)**. Instances where the GOLD label was intelligible but the prediction was unintelligible were counted as **False Positives (FP)**. Finally, instances where both the GOLD label and the prediction were intelligible were counted as **True Negatives (TN)**.

Precision measures the proportion of predicted unintelligible pronunciations that were also judged unintelligible by human annotators, while recall measures the proportion of truly unintelligible pronunciations correctly identified by the model.

$$\text{Precision} = \frac{TP}{TP + FP}$$

$$\text{Recall} = \frac{TP}{TP + FN}$$

The F1 score combines precision and recall into a single measure by weighting them equally:

$$F1 = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

In addition, the F0.5 score was used, which places greater emphasis on precision than recall:

$$F_{0.5} = (1 + 0.5^2) \cdot \frac{\text{Precision} \cdot \text{Recall}}{(0.5^2 \cdot \text{Precision}) + \text{Recall}}$$

In the context of pronunciation assessment, false positive predictions were considered more problematic than false negatives. Incorrectly classifying an intelligible pronunciation as unintelligible was therefore treated as a more severe error than failing to detect an unintelligible pronunciation. For this reason, F0.5 was used as the primary evaluation metric throughout the study.

3.7 IPA model comparison

Model performance was evaluated by strictly comparing IPA token sequences. Each model prediction was compared against all acceptable gold-standard pronunciations for the corresponding target word (which can be found in Appendix B). To be considered correct, the model's output had to be exactly the same. Predictions then received a binary score: 1 if the prediction satisfied the comparison criteria, and 0 otherwise.

These binary prediction scores were subsequently compared against the human intelligibility annotations using the evaluation metrics described above.

All the models were compared using the DEV set. The best-performing one was also evaluated on the TEST set. The latter was done to provide a baseline comparison between one model and an ensemble model.

3.8 Feature-Based Analysis

In addition to direct IPA comparison, the transcriptions were also evaluated using simplified phonological feature representations. This approach was inspired by distinctive feature theory (Jakobson et al., 1952; Chomsky and Halle, 1968; Ladefoged and Maddieson, 1996), in which speech sounds are analyzed as bundles of contrastive properties. The feature inventory used in this study was manually simplified and tailored specifically to the pronunciation assessment task. The simplification was necessary because many established feature systems contain a large number of distinctions that were not relevant to the present analysis and would have introduced unnecessary complexity.

The features are split into consonantal and vocalic features for clarity and interpretability. First, I will present the consonantal features: the feature *place* encodes the primary location of articulation for consonants, such as bilabial, alveolar, or velar. Place of articulation is one of the core dimensions in phonetics because it captures where constriction occurs in the vocal tract during consonant production Ladefoged and Johnson (2014). PanPhon encodes place distinctions using combinations of articulatory features such as [labial], [coronal], and [dorsal] Mortensen et al. (2016). In the present study, these distinctions are collapsed into broader categorical labels because the aim is not to model allophonic detail but rather to capture major structural contrasts between segments.

The feature *manner* describes how airflow is modified during consonant production, distinguishing categories such as stops/plosives[4] , fricatives, nasals, trills, and approximants. Manner contrasts are fundamental in both traditional phonology and computational feature systems because they encode differences in the degree of constriction and sonority (Ladefoged and Johnson, 2014). PanPhon represents these distinctions through combinations of features including [±continuant], [±sonorant], and [±nasal] (Mortensen et al., 2016). The current study simplifies these into interpretable categorical manner labels, which the author considers to be sufficient for modeling broad articulatory similarity without unnecessary theoretical complexity.

The feature *voice* captures whether the vocal folds vibrate during articulation. Voicing contrasts such as /p/ versus /b/ or /s/ versus /z/ are among the most cross-linguistically robust phonological distinctions and are represented nearly all feature theories (Chomsky and Halle, 1968; Hayes, 2009), .

[4]Some sources distinguish between *stops* and *plosives*, while others use the terms interchangeably. Broadly, *stop* refers to consonants produced with complete closure of the vocal tract, whereas *plosive* may refer more specifically to oral stops released with an audible burst of air (Ladefoged and Maddieson, 1996). Under this distinction, nasals may also be considered stops because airflow is blocked orally, despite continued nasal airflow. However, many introductory phonetics texts use the terms synonymously when referring to oral stop consonants such as /p, t, k, b, d, g/ (Clark and Yallop, 1995; Ladefoged and Johnson, 2014). Since the present study focuses on broad articulatory categories relevant to intelligibility and computational modeling rather than fine-grained theoretical distinctions, no functional distinction between the terms is made here. This simplification also reflects the study's broader methodological approach, which prioritizes interpretability and reduced feature complexity over exhaustive phonetic detail.

The feature *asp* represents aspiration, that is, the presence of a period of delayed voicing following stop release (Ladefoged and Johnson, 2014). Aspiration is particularly important in languages like English, where contrasts such as /p/ versus /p^h/ are phonemic. PanPhon models aspiration through laryngeal features such as [spread glottis] (Mortensen et al., 2016). In the present feature system, aspiration is simplified to a binary contrast because only the presence or absence of aspiration is relevant for the intended analysis.

The feature *nasal* identifies whether airflow passes through the nasal cavity during articulation Ladefoged and Johnson (2014). Nasality is commonly treated as a distinctive phonological feature in feature-based phonological theory (Hayes, 2009).

The feature *lateral* distinguishes lateral consonants such as /l/, where airflow escapes around the sides of the tongue, from central consonants, where airflow passes over the center of the tongue (Ladefoged and Johnson, 2014).

The feature *rhotic* identifies rhotic consonants such as trills, taps, or approximant /r/-sounds. Rhoticity is often difficult to model using purely articulatory dimensions because rhotics are phonetically heterogeneous languages (Ladefoged and Johnson, 2014). For example, rhotic sounds may be realized as trills, taps, retroflex approximants, or uvular fricatives depending on the language and dialect. Despite this variation, rhotics frequently behave as a coherent phonological class. [5]

For vowels, the feature *height* represents the vertical tongue position during articulation and distinguishes high, mid, and low vowels (Ladefoged and Johnson, 2014). Vowel height is one of the primary dimensions used in vowel classification and strongly correlates with acoustic formant structure.

The feature *backness* captures the horizontal tongue position in vowel production, distinguishing front, central, and back vowels.

The feature *round* indicates whether the lips are rounded during vowel production. Lip rounding is acoustically salient (Ladefoged and Johnson, 2014).

The feature *long* represents vowel lengths, distinguishing long and short vowels. This is a suprasegmental feature, meaning that the phoneme itself does not encode this; rather, it is encoded with ":" (Ladefoged and Johnson, 2014). Following Jenkins (2000, 2002), where length is considered to be phonemic in this study. [6]

Finally, the feature *diph* identifies whether a vowel consists of a diphthong rather than a monophthong. Unlike monophthongs, diphthongs involve movement between two vowel positions during articulation (Ladefoged and Johnson, 2014). In the present study, diphthongs are marked explicitly because this provides a simple way to distinguish them from monophthongs without modeling detailed vowel movement.

Figure A2 in Appendix A provides an example of a feature bundle representation.

Tables A1 and A2 in Appendix A show the mapping from IPA to features.

IPA segments were converted into feature bundles using PanPhon (Mortensen et al., 2016) as the basis for feature extraction. PanPhon represents IPA segments as binary distinctive-feature vectors, following traditional feature-based phonological theory.

[5]Note that vowels can also be rhotic, thus rhotasized vowels were "opened" to a vowel and a rhotic consonant to ease the analysis. For example, *ɚ" becomes "ə" and "r".

[6]Note that phonologists argue that English does not possess fully phonemic vowel length in the same way as languages such as Finnish or Japanese. Descriptions of English phonology often analyze English vowel contrasts as involving both vowel quality and duration rather than pure length distinctions alone. Wells (1982) notes that vowel duration varies according to phonetic environment. In contrast, Giegerich (1992) discusses English vowel length in relation to syllable structure and timing positions rather than as an entirely independent feature.

These vectors were subsequently mapped onto the smaller feature system used in the present study.

The gold transcriptions and the model predictions were converted into these feature representations. As in the IPA-level analysis, each prediction was compared against all acceptable gold-standard pronunciations and the candidate with the fewest mismatches was retained. Segments occupying the same position in the gold and predicted sequences were then compared feature-by-feature. For consonants, place of articulation, manner of articulation, and voicing were required to match exactly, while aspiration, nasality, and laterality were enforced only when present in the gold segment. For vowels, height and backness were required to match, whereas rounding and length were only evaluated when specified by the gold segment. Segment insertions and deletions were treated as errors.

In addition to strict feature matching, a feature relaxation analysis was conducted. In this procedure, one phonological feature category at a time was ignored during comparison while all remaining features remained constrained. This made it possible to estimate the extent to which individual phonological dimensions contributed to overall mismatch rates and model performance.

Aligned phoneme pairs were additionally used to extract the most frequent segment substitution patterns across models.

3.9 Creating the Ensemble model

3.9.1 Features for Ensemble model

The ensemble models combined model-level outputs with phonological error information extracted from feature-based comparisons. Table 3.2 summarizes the features used in the ensemble models. A more detailed table explaining each feature can be found in Appendix A. Segment-level error counts, phoneme-class error counts, consonantal and vocalic feature error counts, derived error measures, and normalized error-rate features were extracted separately for each of the five speech-to-IPA models, as indicated by the multiplier of five in the count column.

3.9.2 Ensemble models

Several classification algorithms were evaluated as potential ensemble models. These included Logistic Regression, a linear and interpretable baseline classifier; a linear Support Vector Machine (SVM), which is well suited to high-dimensional feature spaces; Random Forest, an ensemble of decision trees capable of modeling non-linear relationships; and XGBoost, a gradient-boosting algorithm that constructs an ensemble of decision trees sequentially to improve predictive performance (James et al., 2023; Breiman, 2001; Chen and Guestrin, 2016).

The models were selected to represent a range of classification approaches, from simple linear methods to more flexible non-linear ensemble techniques. This allowed assessment of whether linear decision boundaries could adequately capture the relationship between phonological error patterns and intelligibility or whether more complex interactions between features were required.

All models were trained using the same feature set (see Table 3.2). The target variable was the human intelligibility label, coded as intelligible (1) or unintelligible

Table 3.2: Features used for ensemble models

Category	**Feature names**	**Feature type**	**Count**
Lexical feature	`word`	Categorical	1
Basic model output and alignment	• `binary` • `n_gold_segments` • `n_pred_segments` • `length_diff`	Binary; integer counts	$4 \times 5 = 20$
Segment-level error counts	• `segment_errors` • `missing` • `extra`	Integer counts	$3 \times 5 = 15$
Phoneme-class error counts	• `consonant_errors` • `vowel_errors`	Integer counts	$2 \times 5 = 10$
Consonantal feature error counts	• `aspiration_errors` • `voicing_errors` • `place_errors` • `manner_errors`	Integer counts	$4 \times 5 = 20$
Vocalic feature error counts	• `height_errors` • `backness_errors` • `round_errors` • `long_errors` • `diph_errors`	Integer counts	$5 \times 5 = 25$
Normalized error-rate features	• `segment_errors_rate` • `missing_rate` • `extra_rate` • `consonant_errors_rate` • `vowel_errors_rate` • `aspiration_errors_rate` • `voicing_errors_rate` • `place_errors_rate` • `manner_errors_rate` • `height_errors_rate` • `backness_errors_rate` • `round_errors_rate` • `long_errors_rate` • `diph_errors_rate` • `length_diff_rate`	Continuous rates	$15 \times 5 = 75$
Derived model-specific features	• `pred_gold_ratio` • `total_feature_errors` • `consonant_minus_vowel_errors`	Continuous ratio; integer count; signed integer difference	$3 \times 5 = 15$
Cross-model vote summary features	• `vote_sum` • `vote_mean` • `vote_std` • `all_models_agree` • `n_models_positive` • `n_models_negative` • `vote_entropy`	Integer counts; continuous summary statistics; binary agreement indicator; entropy score	7
Pairwise model agreement features	• `agree_IPA-Whisper_MultiIPA` • `agree_IPA-Whisper_W2V2-LV60` • `agree_IPA-Whisper_W2V2-XLSR` • `agree_IPA-Whisper_Allosaurus` • `agree_MultiIPA_W2V2-LV60` • `agree_MultiIPA_W2V2-XLSR` • `agree_MultiIPA_Allosaurus` • `agree_W2V2-LV60_W2V2-XLSR` • `agree_W2V2-LV60_Allosaurus` • `agree_W2V2-XLSR_Allosaurus`	Binary agreement indicators	10
Total features			**198**

(0). The categorical word identity feature was one-hot encoded for Logistic Regression, SVM, Random Forest, and XGBoost. Numerical features were standardized for the linear models (Logistic Regression and SVM), while tree-based models used unscaled features.

Model performance was evaluated using nested cross-validation to reduce overfitting and obtain a more reliable estimate of generalization performance. In the outer loop, the DEV data were divided into five folds. In each iteration, four folds were used for training, and one fold was reserved for testing. Within the training portion of each outer fold, an inner cross-validation procedure was used to tune hyperparameters via randomized search. This procedure allowed model hyperparameters to be optimized using only the training data of each fold, while the outer test fold remained unseen until final evaluation.

For each model, the best parameters were found using `RandomizedSearchCV` and optimized for F0.5, since precision was considered more important than recall. 5-fold stratified cross-validation was used during parameter optimization.

All models were implemented using scikit-learn (Pedregosa et al., 2011), except XGBoost, which used the XGBoost library (Chen and Guestrin, 2016).

The model that performed best was then trained on the complete DEV set and tested on the TEST set.

3.10 Error Analysis

To better understand the behavior of the best-performing individual speech-to-IPA model and the final ensemble model, an error analysis was conducted on the test set. The analysis focused on identifying systematic patterns in model errors beyond the aggregate evaluation metrics.

First, confusion matrices were examined for both models. This provided an overview of the distribution of true positives, true negatives, false positives, and false negatives, enabling comparison of the classification errors made by each system.

Second, error rates were analyzed at the word level. For each target word, false-positive and false-negative rates were calculated separately. This made it possible to identify words that were consistently misclassified and to determine whether particular lexical items posed greater difficulty for the models.

Finally, the relationship between classification errors and phonological properties of the target words was investigated. The predictors included aspiration, voicing, diphthongs, vowel length, rhoticity, laterality, vowel quality, and segment counts (see Table 3.3). These predictors were derived automatically from the gold-standard pronunciations described in Section 3.8. Linear regression models were then used to examine whether these properties were associated with overall error rates, false-positive rates, or false-negative rates. Predictors with p-values below .05 were considered statistically significant.

Finally, a qualitative inspection of model outputs was conducted. A sample of 100 recordings from the TEST set was selected based on 50 false positive and 50 false negative classifications made by IPA-Whisper. The recordings were examined manually, and the raw outputs of all speech-to-IPA models were compared in order to identify recurring transcription patterns, model-specific error types, and qualitative differences between the systems. This analysis was intended to complement the

quantitative findings and provide additional insight into the types of pronunciation variation captured by the models.

Together, these analyses provide a more detailed understanding of model behavior and complement the overall performance measures reported earlier.

Table 3.3: Predictors included in the word-level error analyses.

Predictor	**Description**
aspiration	Presence of aspiration
voicing	Presence of voicing contrasts
has_diphthong	Presence of diphthongs
has_long_vowel	Presence of long vowels
has_rhotic	Presence of rhotic segments
has_lateral	Presence of lateral segments
contains_round_vowel	Presence of rounded vowels
contains_back_vowel	Presence of back vowels
contains_open_vowel	Presence of open vowels
n_consonants	Number of consonants
n_vowels	Number of vowels

Chapter 4

Speech-to-IPA Model Comparison on DEV

Here I will present the results of the IPA models. These results are on the development set.

Table 4.1: Baseline: IPA comparison

Model	Metric	Intelligible Class	Unintelligible Class
IPA-Whisper	Precision	0.869	0.500
	Recall	0.266	0.938
	F1	0.407	0.655
	F0.5	0.598	0.552
MultIPA	Precision	0.882	0.452
	Recall	0.072	0.987
	F1	0.134	0.620
	F0.5	0.273	0.507
Wav2Vec2Phoneme - LV-60	Precision	0.848	0.488
	Recall	0.233	0.946
	F1	0.365	0.644
	F0.5	0.555	0.540
Wav2Vec2Phoneme - XLSR-53	Precision	0.859	0.491
	Recall	0.239	0.949
	F1	0.374	0.648
	F0.5	0.566	0.544
Allosaurus	Precision	0.666	0.437
	Recall	0.001	0.994
	F1	0.017	0.607
	F0.5	0.042	0.492

The results of the baseline comparison are in Table 4.1. Overall, there is a clear pattern of high recall and low precision for the unintelligible class and the opposite for the intelligible class.

IPA-Whisper achieved the best overall performance for the intelligible class. Both

Wav2Vec2Phoneme models were closely followed, with the multilingual version (XLSR) performing slightly better. The model that performs the worst overall is Allosaurus.

4.1 Word-level performance

Table 4.2 shows the word-level performance. Note that this is the overall performance of all models combined. The ones highlighted, the models never predict correctly. There are 18 such instances, and all of them are misclassified as unintelligible. Out of those 18 items, 16 start with *p*. This pattern is revisited in Chapter 5, where aspiration-related mismatches emerge as one of the most common sources of transcription disagreement.

The high rate of misclassifying intelligible items as unintelligible coincides with the low recall of the intelligible class.

Table 4.2: Word-level performance metrics by intelligibility class. In = intelligible; Un = unintelligible.

	Precision		Recall		F1		F0.5	
Word	In	Un	In	Un	In	Un	In	Un
bard	0.706	0.758	0.218	0.964	0.333	0.849	0.488	0.792
bay	0.878	0.265	0.465	0.750	0.608	0.392	0.745	0.305
bear	0.600	0.608	0.433	0.752	0.503	0.672	0.557	0.632
bee	0.947	0.105	0.100	0.950	0.181	0.189	0.352	0.128
beer	0.840	0.476	0.191	0.953	0.311	0.635	0.500	0.529
bird	0.370	0.663	0.154	0.864	0.217	0.750	0.289	0.695
board	0.326	0.722	0.273	0.770	0.297	0.746	0.314	0.731
boat	0.735	0.409	0.400	0.743	0.518	0.528	0.630	0.450
boot	0.868	0.480	0.472	0.871	0.611	0.619	0.743	0.528
bout	0.792	0.680	0.447	0.909	0.571	0.778	0.686	0.716
boy	0.940	0.225	0.479	0.833	0.635	0.355	0.788	0.264
buy	0.848	0.260	0.542	0.625	0.661	0.368	0.762	0.295
caught	0.000	0.769	0.000	1.000	0.000	0.870	0.000	0.806
city	0.877	0.323	0.357	0.860	0.508	0.470	0.679	0.369
cot	0.000	0.342	0.000	1.000	0.000	0.510	0.000	0.394
daft	0.829	0.379	0.450	0.783	0.583	0.511	0.709	0.423
dance	0.877	0.265	0.427	0.775	0.574	0.395	0.724	0.305
doll	0.689	0.434	0.365	0.747	0.477	0.549	0.585	0.474
fair	0.679	0.500	0.190	0.900	0.297	0.643	0.448	0.549
farther	1.000	0.618	0.113	1.000	0.202	0.764	0.388	0.669
father	0.638	0.451	0.322	0.753	0.428	0.564	0.533	0.490
fern	1.000	0.412	0.070	1.000	0.130	0.584	0.272	0.467
fir	0.833	0.620	0.067	0.991	0.123	0.763	0.253	0.670
fur	0.632	0.614	0.150	0.939	0.242	0.742	0.385	0.659
half	0.971	0.369	0.252	0.983	0.400	0.536	0.618	0.421
hat	0.951	0.230	0.473	0.867	0.632	0.364	0.791	0.270
knows	0.700	0.464	0.318	0.812	0.438	0.591	0.565	0.508
mate	0.849	0.214	0.413	0.686	0.556	0.327	0.701	0.248

Continued on next page

Table 4.2: Word-level performance metrics by intelligibility class (continued).

	Precision		Recall		F1		F0.5	
Word	In	Un	In	Un	In	Un	In	Un
meat	0.778	0.405	0.183	0.914	0.296	0.561	0.471	0.456
meet	1.000	0.265	0.234	1.000	0.380	0.419	0.605	0.311
nose	0.893	0.235	0.432	0.771	0.583	0.360	0.736	0.273
pat	0.000	0.333	0.000	1.000	0.000	0.500	0.000	0.385
Paul	0.000	0.641	0.000	1.000	0.000	0.781	0.000	0.691
pause	0.000	0.703	0.000	1.000	0.000	0.825	0.000	0.747
paw	1.000	0.771	0.044	1.000	0.085	0.871	0.189	0.808
paws	0.000	0.811	0.000	1.000	0.000	0.896	0.000	0.843
pet	0.000	0.179	0.000	1.000	0.000	0.304	0.000	0.215
pit	0.000	0.282	0.000	1.000	0.000	0.440	0.000	0.329
plate	0.833	0.306	0.231	0.880	0.361	0.454	0.547	0.351
pole	0.000	0.447	0.000	1.000	0.000	0.618	0.000	0.503
pool	0.000	0.525	0.000	1.000	0.000	0.689	0.000	0.580
poor	0.000	0.684	0.000	1.000	0.000	0.812	0.000	0.730
pore	0.000	0.459	0.000	1.000	0.000	0.630	0.000	0.515
pot	0.000	0.300	0.000	1.000	0.000	0.462	0.000	0.349
pour	0.000	0.526	0.000	1.000	0.000	0.690	0.000	0.581
pull	0.000	0.514	0.000	1.000	0.000	0.679	0.000	0.569
put	0.000	0.308	0.000	1.000	0.000	0.471	0.000	0.357
putt	0.000	0.769	0.000	1.000	0.000	0.870	0.000	0.806
seedy	0.714	0.655	0.200	0.950	0.312	0.776	0.472	0.699
tide	1.000	0.612	0.027	1.000	0.052	0.759	0.120	0.663
tied	0.000	0.630	0.000	0.992	0.000	0.770	0.000	0.679
wait	0.827	0.208	0.506	0.550	0.628	0.301	0.734	0.237
weight	0.901	0.263	0.529	0.743	0.667	0.388	0.790	0.302

Chapter 5

Feature-based analysis

In this chapter, I will discuss the results of the feature-based analysis. I will start by looking at error rates per model in Section 5.1. I will look into the most common phoneme substitutions (in Section 5.2) and, lastly, a feature relaxation analysis (in Section 5.3 where each of my features is "relaxed," meaning one-by-one each feature is set as indifferent). All of the analyses in this section were done on the DEV set.

5.1 Error rates per model

Table 5.1 shows the summary of model performance on the segmental level (i.e., phoneme-level). Segment sequences were compared position by position against the best-matching gold-standard pronunciation. Segment substitutions were counted as errors, while segments present only in the gold transcription were classified as missing, and segments present only in the model prediction were classified as extra[1].

All models produce similar numbers of segments, meaning the errors are more likely due to substitution rather than to prediction length. Allosaurus has the most errors, with over 5200, which is almost twice as many as IPA-Whisper, the model with the fewest errors. Allosaurus also has the highest rates for missing and extra segments. This poor performance is not surprising considering what was shown earlier in Table 4.1

Table 5.1: Summary of model performance on the segmental level

Model	segments	errors	missing	extra	accuracy
IPA-Whisper	6237	2724	370	292	0.60
MultIPA	6223	4006	301	754	0.48
W2V2P - LV-60	6209	3005	457	240	0.58
W2V2P - XLSR-5	6190	2876	461	265	0.59
Allosaurus	6166	5218	826	946	0.46

Figure 5.1 shows the error rates of the models per intelligibility class. Note that

[1]For example, if the gold transcription for *bird* is /bɜrd/ and the model predicts /bɜd/, the comparison proceeds position by position. The first two segments (/b/ and /ɜ/) are counted as correct, the third position compares /r/ with /d/ and is counted as a substitution error, and the final segment /d/ in the gold transcription is counted as missing. Thus, the prediction contributes one substitution error and one missing segment.

this figure shows the mismatch between the model's output and the GOLD IPAs, and the results are presented with respect to the intelligibility classes. The error rates are present with respect to the features introduced in Section 3.8 of Chapter 3. There is one extra feature here: type, which is a binary feature of whether the segment is a consonant or a vowel.

If a model were to perform perfectly, the right-hand heatmap (intelligible pronunciations) would contain error rates close to zero across all feature categories. However, this is not observed. Even for pronunciations judged intelligible by human annotators, the models exhibit a non-negligible number of mismatches with the gold-standard transcriptions. Several factors may contribute to these discrepancies. First, the gold-standard IPA representations cannot capture all possible intelligible realizations of a word. Second, some pronunciations may differ from the gold-standard transcription while remaining intelligible to listeners. Finally, the speech-to-IPA systems themselves introduce transcription errors. Consequently, the error rates observed for unintelligible pronunciations should be interpreted with caution, as they reflect both learner pronunciation differences and limitations of the transcription models and reference representations.

Unsurprisingly, the intelligible and the unintelligible classes show the most errors for Allosaurus. IPA-Whisper, on the other hand, has the overall lowest error rates, which aligns with the best performance discussed in the previous chapter.

There are also some general patterns, such as all the models struggling with vowel length in both classes. The most mistakes made in the unintelligible class are related to vowels: roundness, backness, height, length, and diphthongs. Compared to the slightly lower error rates in the intelligible class, one can say that the students likely make quite a few vowel errors. The consonantal features are easier for both the students and the models, given the lower error rates in both classes.

The consistently low error rates for type indicate that the models rarely confuse vowels and consonants. Even when transcriptions differ from the gold standard, the predicted segment is usually of the same broad class as the target segment. This finding is also reassuring regarding the alignment procedure. If alignment errors frequently caused substantial shifts in segment correspondence, higher error rates for type would be expected. Instead, the low rates suggest that the overall consonant-vowel structure is largely preserved and that alignment-induced error propagation is relatively limited.

5.2 Most common substitutions

Table 5.2 shows the most common phoneme substitutions in all of the models. The TOP 3 all concern the aspiration of word-initial plosives. These patterns suggest that the models often correctly identified the place and manner of articulation of consonants while failing to preserve aspiration contrasts. Interestingly, the (by count) most common substitution error $p^h \rightarrow p$ relates to the words with the most errors discussed in the previous chapter (see Table 4.2). That list showed that the models failed most often with words starting with the voiceless bilabial plosives.

Several additional recurrent error patterns were also observed. Voicing confusions were common, particularly *b*→*p*, *d*→*t*, and *t*→*d*, indicating instability in the representation of laryngeal contrasts (i.e., voicing contrast) more generally. Fricative devoicing was also frequent, most notably z→s. In addition, several vowel substitutions reflected differences in vowel quality and diphthongization, such as *ei*→*eɪ*. Taken together, these

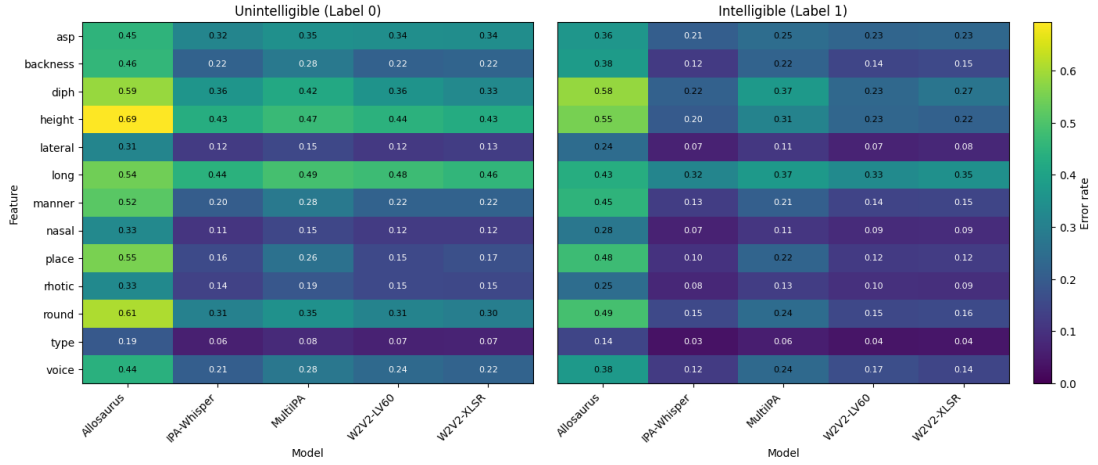


Figure 5.1: Error rates of the models per intelligibility class

findings indicate that the errors produced by the models were not random but instead reflected recurring phonological tendencies.

Given the prominence of aspiration-related substitutions, an additional analysis was conducted in which aspiration-only mismatches were ignored. In this relaxed evaluation, aspirated and unaspirated consonants were treated as equivalent whenever all other consonantal features, including place, manner, and voicing, matched. The purpose of this analysis was to determine the extent to which aspiration alone contributed to overall model errors.

Table 5.2: Most common phoneme substitutions across all models.

Gold	Predicted	Count	Gold Total	Percentage
p^h	p	1932	3085	62.63%
k^h	k	322	381	84.51%
t^h	t	304	362	83.98%
ei	eɪ	293	738	39.70%
p^h	b	275	3085	8.91%
b	p	250	673	37.15%
d	t	248	666	37.24%
t	d	238	1027	23.17%
z	s	232	449	51.67%
f	s	224	638	35.11%

5.3 Feature Relaxation analysis

To further examine which phonological dimensions contributed most strongly to model mismatches, each phonological feature was relaxed independently while all other features remained constrained. The results are shown in Table 5.3. Only features that affected model accuracy are included in the table; the complete results can be found in Table A4 in Appendix A.

The *Gain* column represents the increase in exact-match word accuracy after ig-

noring a given phonological feature. The *Errors* column, in contrast, reports the mean number of mismatches per word attributable to that feature. For example, an aspiration error value of 0.36 indicates that, on average, approximately one aspiration mismatch occurred every three words. Higher error values, therefore, indicate that a feature was a frequent source of disagreement between the predicted and target transcriptions.

The cells highlighted in green indicate large gains in accuracy (≥ 0.06) following feature relaxation. Across all systems, aspiration was the only feature that consistently produced gains of this magnitude. Furthermore, aspiration also showed the highest error rates across nearly all models, indicating that aspiration distinctions are both frequent and highly influential sources of disagreement between predicted and reference phoneme sequences.

Cells highlighted in yellow indicate moderate gains (≥ 0.040). These were observed primarily for voicing and vowel length. For IPA-Whisper and W2V2P-LV60, ignoring vowel length increased intelligible-class accuracy by 0.047, while ignoring voicing produced gains of 0.047 and 0.041, respectively. Although these features contribute to transcription mismatches, their effects are substantially smaller and less consistent than those observed for aspiration.

Features relating to place and manner of articulation, vowel height, backness, and rounding generally produced only minor improvements when relaxed. This suggests that these phonological dimensions contribute relatively little to the transcription disagreements observed in the current dataset. Similarly, Allosaurus exhibited only small gains across all relaxed features, indicating that its comparatively lower performance is unlikely to be driven by any single phonological feature category.

A consistent pattern across models is that the gains obtained through feature relaxation were generally larger for the intelligible class than for the unintelligible class. This asymmetry suggests that many productions judged intelligible by human listeners nevertheless contain phonetic deviations from the canonical IPA transcription. In other words, intelligible pronunciations frequently differ from the reference transcription in terms of aspiration, voicing, or vowel length while remaining understandable. Relaxing these features, therefore, substantially increases agreement with the gold transcription.

By contrast, unintelligible productions appear to contain a broader range of deviations from the reference transcriptions. Although aspiration remains an important source of disagreement, relaxing any single phonological feature typically accounts for a smaller proportion of the total mismatches. This suggests that unintelligible productions are characterized by more widespread phonological divergence, whereas intelligible productions tend to differ from the canonical form in more limited and systematic ways.

Overall, the results identify aspiration as the dominant phonological feature underlying transcription disagreement, with voicing and vowel length playing secondary roles. The consistently larger gains observed for intelligible words further indicate that these feature mismatches do not necessarily correspond to reduced intelligibility, but often reflect phonetic variation that remains understandable to listeners.

Table 5.3: Feature relaxation analysis: accuracy gains after ignoring individual phonological features by intelligibility label - features with effect

		Unintelligible				Intelligible			
Model	Feature	Orig.	Relaxed	Gain	Errors	Orig.	Relaxed	Gain	Errors
W2V2P - XLSR-53	asp	0.102	0.206	+0.104	0.362	0.328	0.464	+0.136	0.241
	voice	0.102	0.125	+0.023	0.103	0.328	0.353	+0.025	0.059
	long	0.102	0.119	+0.017	0.049	0.328	0.361	+0.033	0.054
	place	0.102	0.107	+0.005	0.040	0.328	0.347	+0.019	0.052
	manner	0.102	0.112	+0.010	0.040	0.328	0.346	+0.018	0.031
	height	0.102	0.112	+0.010	0.064	0.328	0.330	+0.002	0.030
	round	0.102	0.103	+0.001	0.005	0.328	0.329	+0.001	0.002
IPA-Whisper	asp	0.094	0.172	+0.078	0.351	0.346	0.472	+0.126	0.237
	voice	0.094	0.141	+0.047	0.129	0.346	0.383	+0.037	0.063
	long	0.094	0.117	+0.023	0.049	0.346	0.393	+0.047	0.071
	place	0.094	0.097	+0.003	0.021	0.346	0.361	+0.015	0.031
	manner	0.094	0.102	+0.008	0.028	0.346	0.357	+0.012	0.020
	height	0.094	0.101	+0.007	0.060	0.346	0.352	+0.006	0.028
	backness	0.094	0.101	+0.007	0.020	0.346	0.348	+0.003	0.007
W2V2P - LV-60	asp	0.093	0.159	+0.066	0.369	0.312	0.426	+0.114	0.243
	voice	0.093	0.121	+0.029	0.135	0.312	0.353	+0.041	0.101
	long	0.093	0.115	+0.022	0.071	0.312	0.359	+0.047	0.069
	place	0.093	0.096	+0.003	0.040	0.312	0.321	+0.009	0.051
	manner	0.093	0.104	+0.012	0.045	0.312	0.326	+0.014	0.027
	height	0.093	0.103	+0.010	0.060	0.312	0.317	+0.004	0.048
	round	0.093	0.097	+0.005	0.008	0.312	0.313	+0.001	0.001
MultiIPA	asp	0.045	0.120	+0.076	0.376	0.117	0.230	+0.113	0.280
	voice	0.045	0.084	+0.039	0.190	0.117	0.153	+0.036	0.163
	long	0.045	0.061	+0.016	0.055	0.117	0.142	+0.025	0.060
	place	0.045	0.050	+0.006	0.062	0.117	0.141	+0.024	0.082
	manner	0.045	0.049	+0.005	0.054	0.117	0.131	+0.014	0.048
	height	0.045	0.049	+0.005	0.054	0.117	0.124	+0.007	0.051
	backness	0.045	0.046	+0.001	0.031	0.117	0.120	+0.003	0.020
Allosaurus	asp	0.022	0.028	+0.006	0.093	0.046	0.054	+0.008	0.072
	voice	0.022	0.029	+0.007	0.062	0.046	0.054	+0.008	0.054
	place	0.022	0.028	+0.006	0.127	0.046	0.059	+0.013	0.133
	long	0.022	0.025	+0.003	0.026	0.046	0.059	+0.012	0.037
	manner	0.022	0.023	+0.001	0.030	0.046	0.050	+0.004	0.038
	height	0.022	0.025	+0.003	0.102	0.046	0.053	+0.007	0.070
	backness	0.022	0.022	+0.000	0.003	0.046	0.048	+0.002	0.008
	round	0.022	0.023	+0.001	0.010	0.046	0.046	+0.000	0.011

Chapter 6

Ensemble model

6.1 Finding the best ensemble model

To recap, four different models were trained and evaluated. These models include logistic regression, Support Vector Machine, Random Forest and XGBoost.

Table 6.1: Results of different ensemble models on DEV

Model	Class	Precision	Recall	F1	F0.5	Accuracy
Logistic Regression	Unintelligible (0)	0.633	0.700	0.665	0.645	0.692
	Intelligible (1)	0.747	0.686	0.715	0.734	
Linear SVM	Unintelligible (0)	0.681	0.588	0.631	0.660	0.700
	Intelligible (1)	0.712	0.787	0.747	0.725	
Random Forest	Unintelligible (0)	0.651	0.688	0.669	0.658	0.703
	Intelligible (1)	0.748	0.715	0.731	0.741	
XGBoost	Unintelligible (0)	0.688	0.651	0.669	0.680	0.719
	Intelligible (1)	0.740	0.771	0.756	0.746	

In Table 6.1 the results of the different models performance on the DEV set. Looking at the F0.5's, the model performance is quite balanced. SVM performs the best for the *unintelligible* class, but the results are quite balanced. XGBoost performs best for the *intelligible* class. Overall accuracy is the best for the XGBoost.

Chapter 7

Results on Test

In this chapter, I will introduce the results of the IPA-Whisper (§ 7.1) and XGBoost (§7.2) on the test set. For both, I will first present the overall performance, then the word-level performance, and lastly compare the overall performance to previous studies (§ 7.3).

7.1 IPA-Whisper

Overall Performance

Table 7.1 shows IPA-Whisper's performance on the TEST data. Its performance slightly improved from the DEV data. The F0.5 jumped from 0.598 to 0.656 for the intelligible class and from 0.552 to 0.572 for the unintelligible class.

Table 7.1: IPA-Whisper results on test set

Metric	Intelligible	Unintelligible
Precision	0.829	0.523
Recall	0.358	0.905
F1-score	0.500	0.663
F0.5-score	0.656	0.572

Word-level Performance

Table 7.2 shows the word-level performance of IPA-Whisper on the test set. Of all the words, the model always incorrectly predicts 14 as unintelligible. These words are highlighted in red. Those in yellow had very low ($\leq$0.100) F0.5. There are 5 such instances, and all of them are mislabeled as unintelligible.

Note that the words that perform badly here also performed badly on the development set (see Table 4.2).

Table 7.2: Word-level IPA-Whisper performance by intelligibility class on TEST

	Precision		Recall		F1		F0.5	
Word	In	Un	In	Un	In	Un	In	Un
bard	0.478	0.743	0.239	0.894	0.319	0.811	0.399	0.769
bay	0.919	0.330	0.535	0.829	0.677	0.472	0.804	0.375
bear	0.727	0.710	0.640	0.786	0.681	0.746	0.708	0.724
bee	1.000	0.106	0.062	1.000	0.118	0.192	0.250	0.129
beer	0.816	0.690	0.705	0.806	0.756	0.744	0.791	0.711
bird	0.594	0.722	0.352	0.875	0.442	0.791	0.522	0.748
board	0.386	0.750	0.386	0.750	0.386	0.750	0.386	0.750
boat	0.768	0.597	0.752	0.617	0.760	0.607	0.765	0.601
boot	0.800	0.539	0.660	0.707	0.723	0.612	0.767	0.566
bout	0.875	0.706	0.500	0.944	0.636	0.808	0.761	0.743
boy	0.963	0.287	0.581	0.885	0.725	0.434	0.851	0.332
buy	0.871	0.432	0.802	0.559	0.835	0.487	0.856	0.452
caught	0.000	0.774	0.000	1.000	0.000	0.872	0.000	0.810
city	0.924	0.429	0.540	0.886	0.682	0.578	0.809	0.478
cot	0.000	0.344	0.000	0.982	0.000	0.509	0.000	0.395
daft	0.936	0.536	0.652	0.900	0.768	0.672	0.861	0.583
dance	0.848	0.370	0.766	0.500	0.805	0.425	0.830	0.390
doll	0.700	0.465	0.442	0.719	0.542	0.564	0.627	0.500
fair	0.807	0.650	0.568	0.855	0.667	0.739	0.744	0.683
farther	0.826	0.650	0.284	0.957	0.422	0.774	0.597	0.694
father	0.716	0.650	0.777	0.574	0.745	0.609	0.727	0.633
fern	0.952	0.463	0.215	0.984	0.351	0.630	0.565	0.518
fir	0.917	0.655	0.180	0.990	0.301	0.788	0.505	0.703
fur	0.875	0.609	0.106	0.989	0.189	0.754	0.357	0.660
half	0.868	0.440	0.536	0.816	0.663	0.571	0.772	0.484
hat	0.915	0.200	0.403	0.800	0.560	0.320	0.730	0.235
knows	0.681	0.574	0.711	0.538	0.696	0.556	0.687	0.566
mate	0.956	0.387	0.694	0.857	0.804	0.533	0.888	0.435
meat	0.833	0.434	0.215	0.933	0.342	0.593	0.529	0.486
meet	0.912	0.275	0.263	0.917	0.408	0.423	0.610	0.320
nose	0.856	0.321	0.712	0.531	0.777	0.400	0.823	0.348
pat	0.000	0.325	0.000	1.000	0.000	0.491	0.000	0.376
paul	0.000	0.631	0.000	0.990	0.000	0.770	0.000	0.680
pause	0.000	0.695	0.000	1.000	0.000	0.820	0.000	0.740
paw	0.444	0.772	0.105	0.958	0.170	0.855	0.270	0.803
paws	0.000	0.789	0.000	1.000	0.000	0.882	0.000	0.824
pet	1.000	0.188	0.008	1.000	0.015	0.316	0.037	0.224
pit	0.500	0.296	0.009	0.979	0.017	0.454	0.041	0.344
plate	0.972	0.538	0.660	0.955	0.787	0.689	0.888	0.590
pole	1.000	0.442	0.011	1.000	0.022	0.613	0.054	0.498

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	Precision		Recall		F1		F0.5	
Word	In	Un	In	Un	In	Un	In	Un
pool	0.000	0.525	0.000	1.000	0.000	0.689	0.000	0.580
poor	0.000	0.677	0.000	1.000	0.000	0.808	0.000	0.724
pore	0.000	0.462	0.000	1.000	0.000	0.632	0.000	0.517
pot	0.000	0.300	0.000	1.000	0.000	0.462	0.000	0.349
pour	0.000	0.538	0.000	1.000	0.000	0.700	0.000	0.593
pull	1.000	0.513	0.013	1.000	0.026	0.678	0.062	0.568
put	0.000	0.300	0.000	1.000	0.000	0.462	0.000	0.349
putt	0.000	0.750	0.000	1.000	0.000	0.857	0.000	0.789
seedy	1.000	0.624	0.033	1.000	0.063	0.769	0.145	0.675
tide	0.000	0.595	0.000	0.989	0.000	0.743	0.000	0.646
tied	1.000	0.630	0.017	1.000	0.034	0.773	0.081	0.680
weight	0.915	0.508	0.835	0.688	0.873	0.584	0.898	0.536

7.2 XGBoost

Overall Performance

The best parameters for the XGBoost trained on the DEV set can be seen in Table A6 in Appendix B. Table 7.3 shows the performance of XGBoost on the test set.

Table 7.3: XGBoost results on test set

Metric	Intelligible	Unintelligible
Precision	0.751	0.701
Recall	0.778	0.669
F1-score	0.764	0.685
F0.5-score	0.756	0.695

Word-level Performance

Table 7.4 shows the word-level performance of XGBoost on the test set. Most words achieved moderate to high performance across both intelligibility classes, although considerable variation can be observed between individual lexical items. The word highlighted in red obtained an F0.5-score of 0.000 for the unintelligible class, indicating that the model failed to correctly identify any instances of that class for the given word. Notably, only one word (*bee*) exhibited this behavior, representing a substantial improvement over the IPA-Whisper baseline, where a much larger number of words received an F0.5-score of 0.000 for the intelligible class. Overall, the results suggest that XGBoost generalizes more consistently across words and is considerably less affected by word-specific classification failures than the baseline IPA-Whisper.

Table 7.4: Word-level XGBoost performance by intelligibility class on TEST

	Precision		Recall		F1		F0.5	
Word	In	Un	In	Un	In	Un	In	Un
bard	0.434	0.783	0.500	0.735	0.465	0.758	0.446	0.773
bay	0.859	0.500	0.866	0.486	0.863	0.493	0.861	0.497
bear	0.631	0.821	0.867	0.548	0.730	0.657	0.667	0.747
bee	0.897	0.000	0.972	0.000	0.933	0.000	0.911	0.000
beer	0.663	0.661	0.784	0.514	0.719	0.578	0.685	0.625
bird	0.519	0.750	0.519	0.750	0.519	0.750	0.519	0.750
board	0.333	0.729	0.341	0.722	0.337	0.726	0.335	0.728
boat	0.738	0.718	0.891	0.467	0.807	0.566	0.764	0.648
boot	0.760	0.844	0.951	0.466	0.845	0.600	0.792	0.726
bout	0.790	0.923	0.914	0.809	0.848	0.862	0.812	0.898
boy	0.870	0.438	0.934	0.269	0.901	0.333	0.882	0.389
buy	0.820	0.700	0.976	0.206	0.891	0.318	0.847	0.473
caught	0.556	0.816	0.278	0.935	0.370	0.871	0.463	0.837
city	0.774	0.485	0.850	0.364	0.810	0.416	0.788	0.455
cot	0.711	0.875	0.981	0.255	0.824	0.394	0.753	0.588
daft	0.870	0.745	0.893	0.700	0.881	0.722	0.874	0.735
dance	0.828	0.458	0.895	0.324	0.860	0.379	0.841	0.423
doll	0.661	0.568	0.800	0.391	0.724	0.463	0.685	0.521
fair	0.750	0.753	0.778	0.724	0.764	0.738	0.755	0.747
farther	0.755	0.748	0.597	0.860	0.667	0.800	0.717	0.768
father	0.654	0.719	0.904	0.338	0.759	0.460	0.692	0.587
fern	0.662	0.481	0.570	0.578	0.613	0.525	0.642	0.497
fir	0.865	0.758	0.525	0.948	0.653	0.843	0.766	0.790
fur	0.684	0.735	0.591	0.806	0.634	0.769	0.663	0.749
half	0.805	0.587	0.827	0.551	0.816	0.568	0.810	0.579
hat	0.868	0.304	0.881	0.280	0.874	0.292	0.870	0.299
knows	0.656	0.733	0.911	0.338	0.763	0.463	0.695	0.595
mate	0.917	0.581	0.895	0.643	0.906	0.610	0.913	0.592
meat	0.766	0.810	0.914	0.567	0.833	0.667	0.791	0.746
meet	0.838	0.625	0.924	0.417	0.879	0.500	0.854	0.568
nose	0.847	0.462	0.888	0.375	0.867	0.414	0.855	0.441
pat	0.681	0.375	0.907	0.115	0.778	0.176	0.716	0.259
paul	0.587	0.723	0.466	0.810	0.519	0.764	0.558	0.739
pause	0.536	0.827	0.638	0.757	0.583	0.790	0.554	0.812
paw	0.333	0.774	0.184	0.883	0.237	0.825	0.287	0.793
paws	0.556	0.811	0.156	0.967	0.244	0.882	0.368	0.838
pet	0.836	0.556	0.969	0.167	0.898	0.256	0.859	0.379
pit	0.735	0.386	0.761	0.354	0.748	0.370	0.740	0.379
plate	0.904	0.739	0.887	0.773	0.895	0.756	0.900	0.746
pole	0.704	0.515	0.432	0.768	0.535	0.616	0.625	0.551
pool	0.733	0.626	0.434	0.857	0.545	0.724	0.645	0.662
poor	0.568	0.754	0.420	0.848	0.483	0.798	0.530	0.771
pore	0.896	0.620	0.512	0.931	0.652	0.744	0.779	0.665

	Precision		Recall		F1		F0.5	
Word	In	Un	In	Un	In	Un	In	Un
pot	0.720	0.471	0.920	0.167	0.808	0.246	0.753	0.345
pour	0.903	0.648	0.389	0.964	0.544	0.775	0.714	0.693
pull	0.550	0.547	0.434	0.658	0.485	0.598	0.522	0.566
put	0.832	0.566	0.795	0.625	0.813	0.594	0.824	0.577
putt	0.500	0.767	0.125	0.958	0.200	0.852	0.312	0.799
seedy	0.641	0.789	0.672	0.765	0.656	0.777	0.647	0.785
tide	0.529	0.761	0.742	0.554	0.617	0.642	0.561	0.708
tied	0.577	0.798	0.707	0.691	0.636	0.740	0.599	0.774
weight	0.883	0.649	0.933	0.500	0.907	0.565	0.893	0.612

7.3 Comparison

Table 7.5 compares the performance of the proposed IPA-Whisper and XGBoost models against previous work. Although all studies address the same binary intelligibility classification task, the models and datasets differ slightly.

Table 7.5: Comparison of performance to previous research

Model	Class	Precision	Recall	F1-score	F0.5-score
IPA-Whisper	Intelligible	0.829	0.358	0.500	0.656
	Unintelligible	0.523	0.905	0.663	0.572
XGBoost	Intelligible	0.751	0.778	0.764	0.756
	Unintelligible	0.701	0.669	0.685	0.695
Kuan (2025)	Intelligible	0.741	0.562	0.654	0.600
	Unintelligible	0.581	0.726	0.616	0.588
Zou (2024)	Intelligible	0.732	0.565	0.638	0.691
	Unintelligible	0.403	0.586	0.477	0.429

The results are compared with those reported by Zou (2024) and Kuan (2025). Zou (2024) evaluated Whisper-based intelligibility classification on the smaller MOOC-2024 dataset containing 2,631 word recordings, while Kuan (2025) investigated continual pretraining of Whisper models on isolated word recognition and evaluated intelligibility classification using a phonetic matching approach on the MOOC-2025 dataset of approximately 10,000 tokens. As the datasets and experimental procedures differ, the results should be interpreted as indicative rather than strictly comparable.

Among the evaluated systems, XGBoost achieved the strongest overall performance. For both classes, the XGBoost achieved the highest F0.5-score, with 0.756 for Intelligible and 0.695 for unintelligible.

The IPA-Whisper baseline exhibited different behavior. It achieved the highest Precision for the intelligible class (0.829) and the highest recall for the unintelligible class (0.905), indicating a conservative classification strategy that rarely labels unintelligible pronunciations as intelligible. However, this came at the cost of lower recall for the intelligible, as well as a lower precision for unintelligible class and consequently lower F1 and F0.5-scores.

Overall, the results demonstrate that combining phonological error features from multiple phoneme recognition systems within a supervised ensemble substantially improves performance over a rule-based approach and compares favorably with previously reported ASR-based methods.

Chapter 8

Error Analysis

8.1 Confusion Matrices

Figures 8.1 and 8.2 show the confusion matrices of IPA-Whisper and XGBoost, respectively.

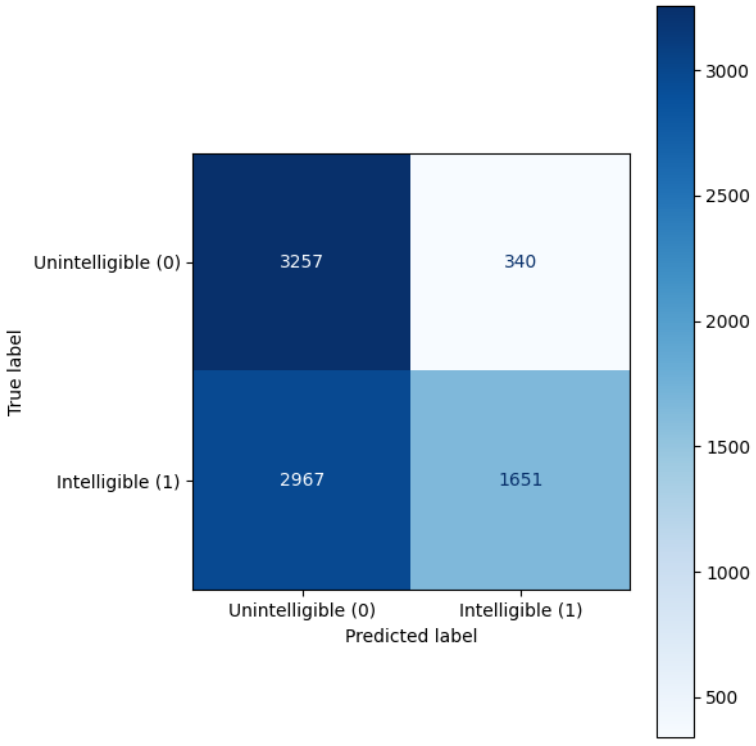


Figure 8.1: IPA-Whisper Confusion Matrix on TEST

For IPA-Whisper, 3,257 unintelligible words were correctly classified, and only 340 unintelligible words were incorrectly classified as intelligible. This corresponds to the very high recall for the unintelligible class (0.905) reported in Table 7.1. However, the model correctly identified only 1,651 intelligible words and misclassified 2,967 as unintelligible. As a result, unintelligible-class precision is much lower (0.523 as reported in Table 7.1).

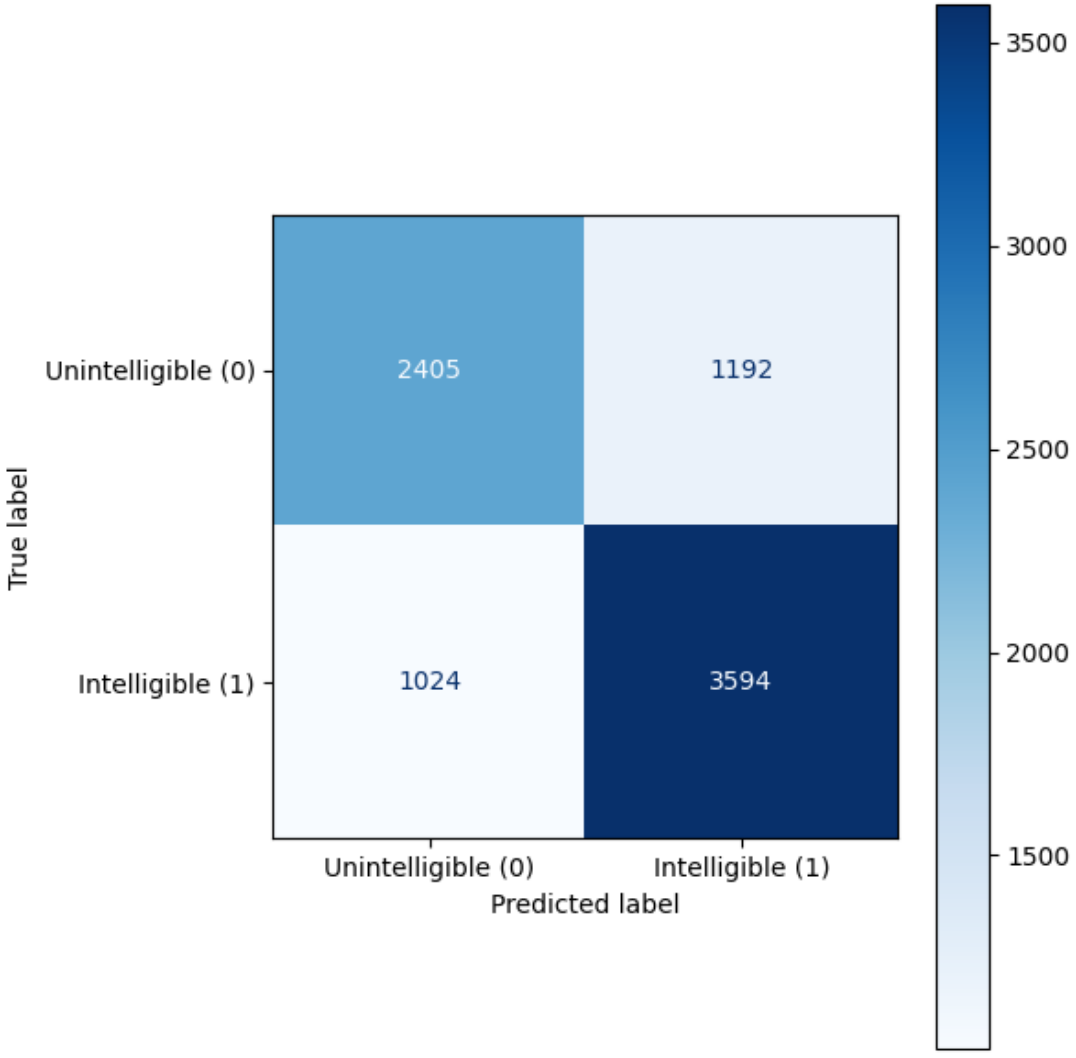


Figure 8.2: XGBoost Confusion Matrix on TEST

In contrast, the XGBoost confusion matrix showed a much more balanced distribution of errors. XGBoost correctly classified 2,405 unintelligible words and 3,594 intelligible words, while producing 1,192 false negatives and 1,024 false positives. Compared with IPA-Whisper, XGBoost reduced the number of intelligible words incorrectly classified as unintelligible from 2,967 to 1,024, a reduction of more than 50%. This improvement came at the cost of more false negatives for the unintelligible class (1192 versus 340).

The difference between the two confusion matrices demonstrates the advantage of the ensemble approach. IPA-Whisper relies on strict phonological matching between the predicted and reference transcriptions. Consequently, many pronunciations that are intelligible to human listeners are nevertheless classified as unintelligible because they contain phonological deviations from the canonical form. By incorporating phonological feature errors, agreement statistics, and outputs from multiple phoneme recognition systems, XGBoost is better able to distinguish between minor phonological variation and genuinely unintelligible pronunciations. This results in a much more balanced trade-off between precision and recall and substantially higher F1-scores for both classes.

8.2 Words with high false positive and false negative rates

IPA-Whisper

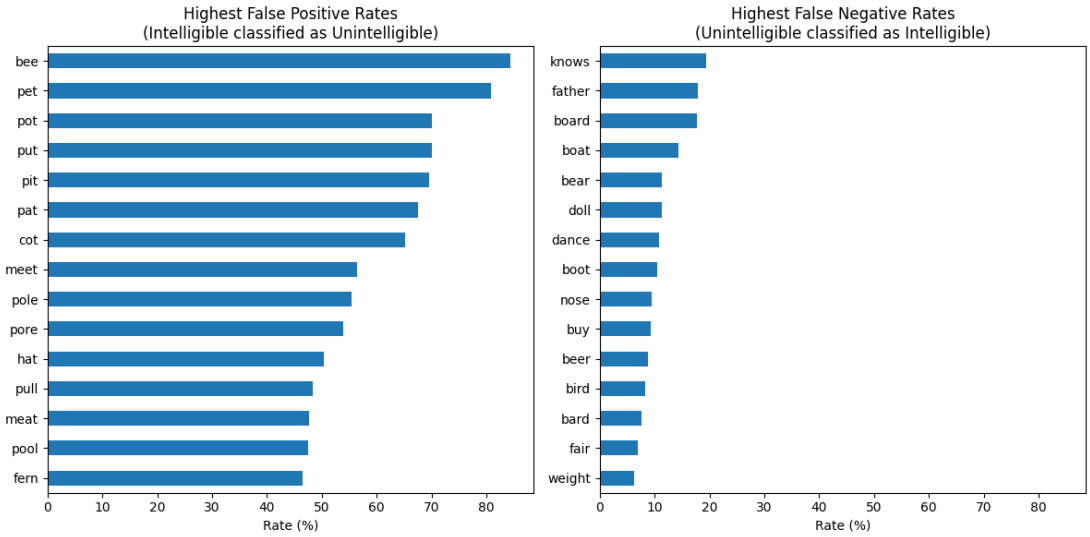


Figure 8.3: IPA-Whisper: Highest FN and FP rates per word

Figure 8.3 shows the words with the highest false positive and false negative rates for IPA-Whisper. As a reminder false positives are intelligible instances mispredicted as unintelligible and false negatives are unintelligible instances misclassified as intelligible. The two error types reveal a clear asymmetry. False positives occurred much more frequently than false negatives. The highest false positive rates were observed for *bee* (85%), *pet* (81%), *pot* (70%), *put* (70%), and *pit* (69%). In other words, these words

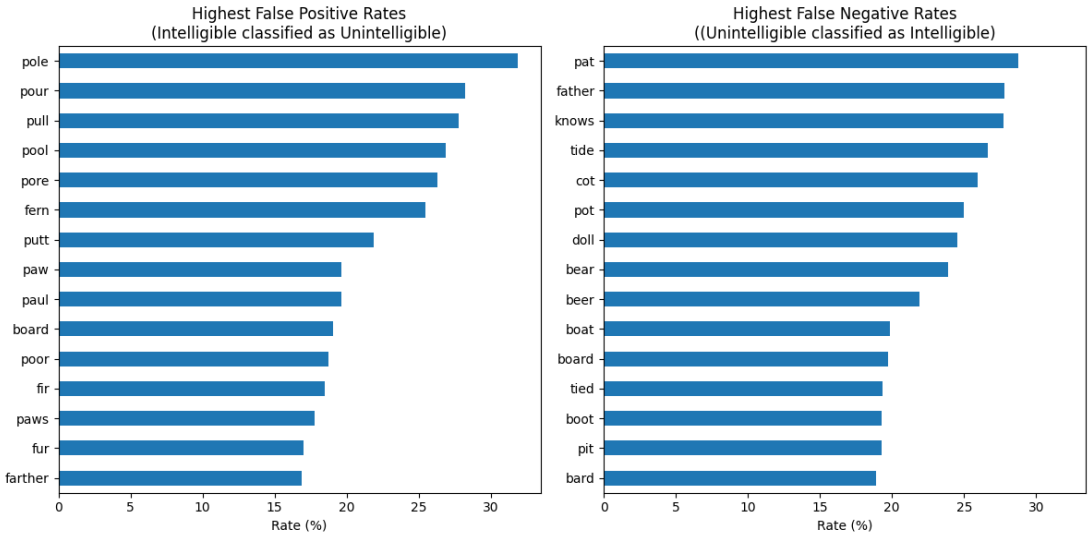


Figure 8.4: XGBoost: Highest FN and FP rates per word

were frequently rejected by the model despite being judged intelligible by human annotators.

False negatives occurred at substantially lower rates. The highest false negative rates were found for *knows* (19%), *father* (18%), *board* (17%), *boat* (14%), and *bear* (11%).

A notable pattern is that many of the words with high false positive rates begin with aspirated voiceless stops and belong to minimal vowel sets, including *pit*, *pet*, *pat*, *put*, and *pot*. Several words containing liquids, such as *pole*, *pore*, *pull*, *pool*, and *fern*, also appear among the most error-prone items. This observation is consistent with the feature-relaxation analysis, which identified aspiration as the single most influential source of disagreement between predicted and reference transcriptions. In particular, the frequent omission of aspiration caused many pronunciations that were intelligible to human listeners to be classified as unintelligible.

Overall, the remaining errors made by IPA-Whisper are highly asymmetric. The model is considerably more likely to underestimate intelligibility than to overestimate it. This suggests that relatively small phonetic deviations, especially those involving aspiration and liquids, are often treated as evidence of unintelligibility even when they do not impair comprehension.

XGBoost

Figure 8.4 shows the words with the highest false positive and false negative rates for the XGBoost classifier. The two error types reveal different patterns.

Several words that were problematic for IPA-Whisper also appear among the most difficult items for XGBoost, suggesting that at least some errors reflect intrinsic properties of the words themselves rather than weaknesses of a particular model.

False positives, where intelligible productions were classified as unintelligible, were most common for *pole* (32%), *pour* (28%), *pull* (28%), *pool* (27%), and *pore* (26%). These words were frequently rejected by the model despite being judged intelligible

by human annotators. As already observed with IPA-Whisper, most of the highest false-positive words contain liquids (/l/ or /r/) and many also contain rounded vowels or aspirated voiceless stops.

False negatives, where unintelligible productions were classified as intelligible, occurred at slightly lower rates. The highest false negative rates were observed for *pat* (29%), *father* (28%), *knows* (28%), *tide* (27%), and *cot* (26%). Unlike the false-positive words, these items do not exhibit a single clear phonological pattern. Some contain back vowels or rhotics, whereas others are characterized by short vowels or voiceless stops. Overall, the false-negative words appear to be more heterogeneous.

A notable pattern is that many words with high false-positive rates contain liquids, particularly /l/ and /r/. The concentration of words such as *pole*, *pour*, *pool*, *pore*, and *pull* among the most error-prone items further suggests that liquids remain difficult for the ensemble classifier despite the improvements over the individual speech-to-IPA models. In contrast, the words with high false-negative rates are more varied and do not cluster around a single phonological property. Overall, the remaining errors made by XGBoost appear to be driven primarily by words containing liquids and other phonological features that allow substantial pronunciation variation while remaining intelligible.

8.3 Phonological Properties Associated with Classification Errors

IPA-Whisper

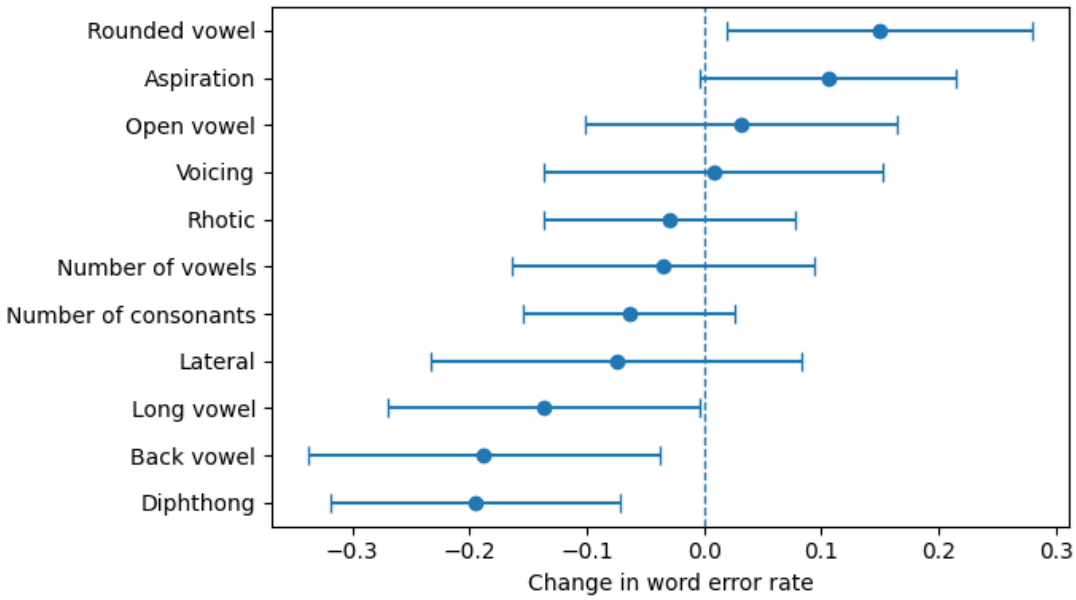


Figure 8.5: IPA-Whisper: Linear Regression Coefficients with the 95% confidence intervals

Similar to IPA-Whisper, a linear regression model was fitted using word-level XGBoost error rates as the dependent variable. Whereas the IPA-Whisper analyses primarily highlighted aspiration and vowel-related properties, the XGBoost analyses allow us to examine which phonological factors remain problematic after combining information from multiple speech-to-IPA systems.

Figure 8.5 visualizes the regression coefficients and their 95% confidence intervals, while the full model statistics are reported in Table 8.1. The model explained a moderate proportion of the variation in word-level error rates (R^2 = 0.476, adjusted R^2 = 0.332, p = .003). The coefficients can be interpreted as the expected change in error rate associated with the presence of a particular phonological property while holding the remaining predictors constant. Positive coefficients indicate increased error rates, whereas negative coefficients indicate reduced error rates. Predictors whose confidence intervals do not cross zero can be interpreted as having statistically reliable effects.

Several vowel-related properties emerged as significant predictors of error rates. As a reminder, predictors with p-values below .05 were considered statistically significant. Words containing rounded vowels showed significantly higher error rates (β = 0.150, p = .026), corresponding to an increase of approximately 15 percentage points relative to words without rounded vowels. In contrast, words containing diphthongs (β = -0.196, p = .003), back vowels (β = -0.188, p = .015), and long vowels (β = -0.137, p = .045) exhibited significantly lower error rates, corresponding to reductions of approximately 20, 19, and 14 percentage points, respectively. No consonant-related features reached significance. Overall, these results suggest that variation in IPA-Whisper error rates is driven primarily by vowel quality rather than consonantal properties.

To determine whether false negatives drove these effects or false positives, separate regression models were fitted for each error type (see Table 8.2).

The FN model explained approximately half of the variation (R^2 = 0.502, adjusted R^2 = 0.365, p = .001). Three predictors reached significance. Aspiration substantially increased the FN rate (β = 0.177, p = .007), indicating that intelligible words containing aspirated consonants were more likely to be misclassified by the model. In contrast, diphthongs (β = -0.217, p = .003) and back vowels (β = -0.177, p = .042) reduced FN rates.

The FP model showed an overall similar fit (R^2 = 0.526, adjusted R^2 = 0.396, p < .001). However, the significant predictors differed. Rounded vowels increased FP rates (β = 0.053, p = .014), indicating that unintelligible words containing rounded vowels were more likely to be incorrectly accepted as intelligible. Aspiration showed the opposite pattern, significantly reducing FP rates (β = -0.071, p < .001). Together with the FN results, this suggests that aspiration makes IPA-Whisper more conservative: words containing aspirated consonants are more likely to be classified as unintelligible regardless of the true label.

These findings are consistent with the word-level analyses presented earlier. Many of the words exhibiting the highest false-negative rates began with aspirated voiceless stops, particularly /p/. Together, the regression and feature-relaxation analyses suggest that aspiration-related variation contributes disproportionately to classification errors, even when the productions remain intelligible to human listeners.

Table 8.1: Linear regression predicting word-level IPA-Whisper error rates from phonological properties. Significant predictors are shown in bold.

Predictor	Coef.	SE	p	95% CI
Intercept	0.703	0.111	$< .001$	[0.479, 0.926]
Aspiration	0.106	0.054	.058	[-0.004, 0.215]
Voicing	0.008	0.072	.911	[-0.136, 0.153]
Diphthong	**-0.196**	**0.061**	**.003**	**[-0.319, -0.072]**
Long vowel	**-0.137**	**0.066**	**.045**	**[-0.270, -0.003]**
Rhotic	-0.030	0.053	.575	[-0.137, 0.077]
Lateral	-0.075	0.078	.344	[-0.233, 0.083]
Rounded vowel	**0.150**	**0.065**	**.026**	**[0.019, 0.280]**
Back vowel	**-0.188**	**0.074**	**.015**	**[-0.338, -0.038]**
Open vowel	0.032	0.066	.634	[-0.101, 0.165]
Number of consonants	-0.064	0.045	.162	[-0.154, 0.027]
Number of vowels	-0.035	0.064	.590	[-0.164, 0.094]
N				52 words
R^2				0.476
Adjusted R^2				0.332
F				3.31 ($p = .003$)

Table 8.2: Linear regression models predicting word-level false-negative and false-positive rates for IPA-Whisper. Significant predictors are shown in bold.

	False Negatives		False Positives	
Predictor	Coef.	p	Coef.	p
Intercept	0.693	$< .001$	0.009	.793
Aspiration	**0.177**	**.007**	**-0.071**	**$< .001$**
Voicing	0.015	.851	-0.007	.749
Diphthong	**-0.217**	**.003**	0.022	.263
Long vowel	-0.129	.093	-0.008	.711
Rhotic	-0.023	.708	-0.007	.668
Lateral	-0.066	.462	-0.009	.717
Rounded vowel	0.097	.193	**0.053**	**.014**
Back vowel	**-0.177**	**.042**	-0.011	.635
Open vowel	0.005	.945	0.026	.210
Number of consonants	-0.085	.104	0.021	.150
Number of vowels	-0.038	.607	0.003	.887
R^2	0.502		0.526	
Adjusted R^2	0.365		0.396	
Model p	.001		$< .001$	
N	52		52	

XGBoost

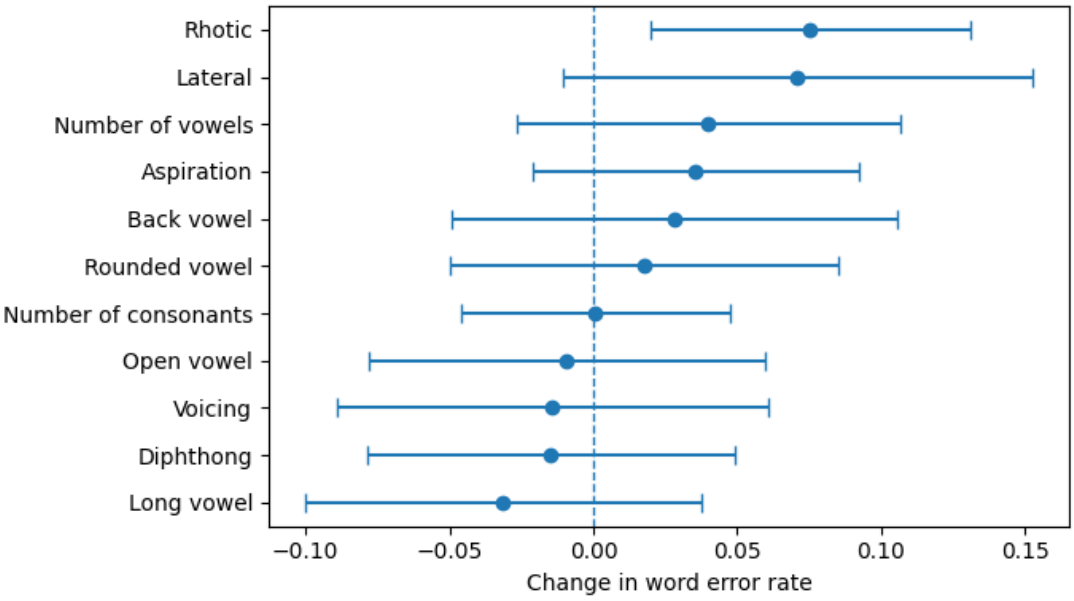


Figure 8.6: XGBoost: Linear regression coefficients with 95% confidence intervals

Table 8.3: Linear regression predicting word-level XGBoost error rates from phonological properties. Significant predictors are shown in bold.

Predictor	Coef.	SE	p	95% CI
Intercept	0.185	0.057	.003	[0.069, 0.300]
Aspiration	0.036	0.028	.210	[-0.021, 0.092]
Voicing	-0.014	0.037	.703	[-0.089, 0.061]
Diphthong	-0.015	0.032	.645	[-0.079, 0.049]
Long vowel	-0.031	0.034	.364	[-0.100, 0.038]
Rhotic	**0.076**	0.027	**.009**	[0.020, 0.131]
Lateral	0.071	0.040	.086	[-0.011, 0.153]
Rounded vowel	0.018	0.033	.603	[-0.050, 0.085]
Back vowel	0.028	0.038	.467	[-0.049, 0.106]
Open vowel	-0.009	0.034	.790	[-0.078, 0.060]
Number of consonants	0.001	0.023	.975	[-0.046, 0.048]
Number of vowels	0.040	0.033	.234	[-0.027, 0.107]
N				52 words
R^2				0.407
Adjusted R^2				0.244
F				2.50 ($p = .017$)

Similar to IPA-Whisper, a linear regression model was fitted using word-level XGBoost error rates as the dependent variable. The model explained a moderate proportion of the variation in error rates ($R^2 = 0.407$, adjusted $R^2 = 0.244$, $p = .017$). As shown in Table 8.3, only the presence of a rhotic was a significant predictor. Words containing rhotics exhibited error rates approximately 7.6 percentage points higher

than words without rhotics. Laterals showed a similar positive tendency, although the effect did not reach significance ($p = .086$). None of the remaining phonological properties significantly predicted overall error rates.

As with IPA-Whisper, separate regression models were fitted for each error type (see Table 8.4).

Table 8.4: Linear regression models predicting word-level false-negative and false-positive rates for XGBoost. Significant predictors are shown in bold.

	False Negatives		False Positives	
Predictor	Coef.	p	Coef.	p
Intercept	0.153	.014	0.032	.476
Aspiration	0.001	.977	0.035	.116
Voicing	0.010	.803	-0.024	.409
Diphthong	-0.019	.573	0.004	.872
Long vowel	**-0.083**	**.025**	0.052	.058
Rhotic	-0.023	.426	**0.098**	**< .001**
Lateral	-0.045	.294	**0.116**	**.001**
Rounded vowel	0.048	.178	-0.030	.252
Back vowel	-0.038	.342	**0.067**	**.031**
Open vowel	0.019	.588	-0.029	.286
Number of consonants	0.003	.906	-0.002	.905
Number of vowels	0.018	.598	0.022	.400
R^2	0.235		0.638	
Adjusted R^2	0.025		0.539	
Model p	.374		$< .001$	
N	52		52	

The false-negative model explained little variation ($R^2 = 0.235$, adjusted $R^2 = 0.025$) and was not statistically significant overall ($p = .374$). Among the predictors, only long vowels reached significance. Words containing long vowels exhibited lower false-negative rates, suggesting that intelligible words containing vowel length contrasts were less likely to be classified as unintelligible. No effects were observed for rhotics, laterals, aspiration, or segment counts.

In contrast, the false-positive model explained a substantial proportion of the variation in word-level false-positive rates ($R^2 = 0.638$, adjusted $R^2 = 0.539$, $p < .001$). Words containing rhotics and laterals showed significantly higher false-positive rates, increasing the expected rate by approximately 10 and 12 percentage points, respectively. Back vowels also emerged as a significant predictor, increasing false-positive rates by roughly 7 percentage points. Long vowels showed a marginal positive effect ($p = .058$). These results suggest that most systematic XGBoost errors stem from unintelligible words containing rhotics, laterals, and back vowels being incorrectly classified as intelligible.

8.4 Qualitative inspection of model outputs

Several systematic tendencies emerged during the qualitative inspection. Overall, IPA-Whisper produced the most plausible English transcriptions. Even when its outputs differed from the gold-standard transcription, the resulting forms often remained

phonetically similar to the target word. For example, intelligible productions of *buy* and *farther* were transcribed as /bˈaɪd/ and /fˈaɪvɚ/, respectively. These outputs contain additional segments or substitutions while still preserving much of the target's phonetic structure. In a smaller number of cases, IPA-Whisper produced substantially different outputs, such as /tˈɪ/ for *pour*. For unintelligible productions, the model often appeared to capture the pronunciation deviations present in the recording. For example, in one production of *boat*, the target diphthong was realized closer to a monophthong by the speaker, which was reflected in the model output /bˈɔt/.

The two Wav2Vec2Phoneme models exhibited broadly similar behavior. In many cases, both models preserved the overall shape of the target word while differing in vowel quality, rhoticity, or aspiration. For example, an intelligible production of *put* was transcribed as /pot/ by W2V2P-LV60 but as /put/ by W2V2P-XLSR. Likewise, an intelligible production of *plate* was rendered as /leɪt/ by W2V2P-LV60 and /lɜːt/ by W2V2P-XLSR, with both models omitting the word-initial /p/. Both models also frequently omitted aspiration, which is consistent with the quantitative analyses showing aspiration as one of the dominant sources of transcription disagreement. Nevertheless, the resulting outputs generally remained recognizably related to the target word.

MultiIPA displayed greater variability than IPA-Whisper and the Wav2Vec2Phoneme models. While many outputs remained phonetically close to the target, the model occasionally introduced segments that are uncommon in English. Examples included productions such as /pa͡tʂ/ for *pat*, /ɬat/ for *plate*, and retroflex realizations such as /bɛɻd/ and /boɻt/ for *bird* and *board*. These outputs remain broadly interpretable but draw from a phonetic inventory that extends beyond the sounds typically represented in English phonemic transcriptions. This behavior is consistent with the model's multilingual training objective. Because MultiIPA was trained to recognize and represent speech across multiple languages, it has been exposed to a substantially broader range of phonetic categories than would be encountered in English alone. Consequently, when faced with acoustically ambiguous pronunciations, the model may select phones that are phonetically plausible but uncommon in English, resulting in transcriptions that appear more multilingual than those produced by IPA-Whisper or the Wav2Vec2Phoneme models.

The most distinctive behavior was observed for Allosaurus. Unlike the other models, Allosaurus frequently produced phones that are not normally associated with English phonology. Examples included uvular fricatives (/ʁ/), retroflex nasals (/ɳ/), uvular nasals (/ɴ/), central vowels (/ɨ/), retroflex fricatives (/ʂ/), and alveolo-palatal affricates (/t͡ɕ/). For example, intelligible productions of *cot* and *nose* were transcribed as /ʁ ɔ ɳ ɴ t͡ɕ/ and /j ɛ ʂ/, respectively. Similarly, productions of *boot*, *city*, and *farther* contained phones such as /ɨ/, /ʂ/, and /x/, which are uncommon in standard English phonological descriptions. These observations are consistent with the quantitative analyses, which showed that Allosaurus exhibited the highest rates of substitution, insertion, and deletion. The model appears to prioritize broad multilingual phonetic coverage rather than English-specific transcription accuracy.

Taken together, the qualitative inspection suggests that the stronger-performing models primarily differ from the gold-standard transcriptions through relatively minor phonetic deviations, such as vowel-quality differences, aspiration omissions, or rhoticity mismatches. In contrast, the weaker-performing models, particularly Allosaurus, more frequently produce qualitatively different phone sequences and draw from substantially broader phonetic inventories. These observations help explain the quantitative findings

presented earlier. The relatively small deviations produced by IPA-Whisper and the Wav2Vec2Phoneme models are consistent with their lower feature-level error rates and stronger intelligibility classification performance, whereas the more divergent outputs produced by Allosaurus align with its elevated insertion, deletion, and substitution rates.

Chapter 9

Discussion

The discussion will start with an analysis of the results, followed by a section on the phonological patterns the models exhibited. A small post hoc test is then conducted to check for lexicality bias in the word list and to assess whether there is any correlation between error rate and lexical frequency. This is followed by a section that answers the research question, a section that discusses the limitations of the current study, and, lastly, a section that proposes ideas for future research.

9.1 Analysis of results

The central finding of the present study is that speech-to-IPA systems capture information relevant to intelligibility, but transcription accuracy alone is an insufficient proxy for intelligibility. Across all evaluated models, substantial disagreement was observed between model outputs and the gold-standard transcriptions, even for pronunciations judged intelligible by human annotators. At the same time, the ensemble classifier substantially outperformed all individual speech-to-IPA systems. Taken together, these findings suggest that intelligibility is better represented as a pattern of interacting phonological cues than as exact agreement with a canonical transcription.

The individual speech-to-IPA models showed considerable variation in performance. IPA-Whisper achieved the strongest results among the individual systems and consistently produced transcriptions that were most similar to the gold-standard IPA representations. However, even IPA-Whisper exhibited substantial disagreement with the reference transcriptions and produced many errors when transcription accuracy was used directly as an intelligibility classifier. This indicates that accurate phonetic transcription and intelligibility, while related, are not identical constructs.

The feature-relaxation analysis provides further support for this interpretation. Across the evaluated models, aspiration, voicing, and vowel-length mismatches frequently contributed to disagreement with the gold-standard transcriptions. However, many of these mismatches occurred in productions that were nevertheless judged intelligible by human annotators. In other words, a pronunciation could deviate substantially from the canonical transcription while remaining understandable to listeners. These findings suggest that direct transcription comparison may overestimate the communicative importance of some phonetic deviations.

This interpretation is consistent with previous work distinguishing pronunciation accuracy from intelligibility. Munro and Derwing (1999) demonstrated that accented speech can remain highly intelligible despite deviating from native-speaker norms.

The present results suggest a similar pattern in automatic pronunciation assessment. While phonological deviations undoubtedly contribute to reduced intelligibility, the relationship is not one-to-one. Consequently, systems that rely exclusively on transcription accuracy may misrepresent the extent to which a pronunciation affects listener understanding.

The superior performance of the ensemble classifier provides additional evidence for this interpretation. Rather than relying on exact transcription agreement, the XGBoost model incorporated information from multiple sources, including segmental errors, feature-level mismatches, model agreement patterns, and broader phonological trends. This allowed the classifier to learn which deviations were associated with unintelligibility and which were comparatively benign. The success of the ensemble, therefore, suggests that intelligibility is best understood as a problem of cue integration rather than direct transcription matching.

9.1.1 Complementary information across models

One of the most notable findings of the present study is that the ensemble classifier substantially outperformed every individual speech-to-IPA system. This improvement suggests that the models capture different aspects of pronunciation rather than simply making the same errors to different degrees.

The feature-based analyses demonstrated that the models differed in their phonological sensitivities, substitution patterns, and error distributions. IPA-Whisper generally produced transcriptions that were closest to the gold-standard IPA forms, whereas the Wav2Vec2Phoneme models often preserved the broader phonological structure while omitting finer phonetic detail such as aspiration. MultiIPA and Allosaurus exhibited different error profiles again, reflecting their multilingual orientation and broader phonetic inventories.

The ensemble classifier was able to exploit these differences. Rather than treating disagreement between models as noise, it used such disagreement as a source of information. A pronunciation that appears problematic to one model may be treated differently by another, and these differences can themselves be informative for intelligibility prediction. Consequently, the ensemble classifier does not merely aggregate multiple transcriptions but instead integrates complementary phonological evidence from different representational levels.

This finding has broader implications for pronunciation assessment. It suggests that intelligibility is unlikely to be adequately represented by any single transcription system. Instead, intelligibility appears to emerge from the interaction of multiple phonological cues, each captured to varying degrees by different models. The success of the ensemble, therefore, supports the view that pronunciation assessment benefits from integrating multiple sources of evidence rather than relying on a single phonetic representation.

9.2 Phonological patterns

This section discusses the specific phonological patterns that emerged during the feature-based and error analyses.

9.2.1 Cue weighting and intelligibility

The phonological patterns observed in the present study can be interpreted through the lens of cue-weighting theory. Cue-weighting approaches to speech perception propose that listeners do not rely equally on all available acoustic cues when identifying speech sounds. Instead, different cues are assigned different perceptual weights depending on how informative and reliable they are for distinguishing linguistic categories (Holt and Lotto, 2006).

Speech sounds are typically characterized by multiple acoustic cues simultaneously. For example, stop consonants may be distinguished by voice-onset time (VOT), burst characteristics, formant transitions, and duration cues. However, listeners do not necessarily use all of these cues to the same extent. Through language experience, listeners learn which cues are most reliable and tend to rely more heavily on those dimensions during speech perception.

A similar pattern emerged in the present study. The feature-relaxation analysis revealed substantial differences between phonological feature categories. Relaxing aspiration and vowel-length distinctions produced some of the largest increases in agreement, while several vowel-related dimensions showed more variable effects across models. This suggests that the evaluated models were not equally sensitive to all dimensions of pronunciation and that some phonological contrasts contributed more strongly to disagreement than others.

Importantly, many of these mismatches occurred in pronunciations that were nevertheless judged intelligible by human annotators. This indicates that phonological deviations do not necessarily have an equal impact on listener understanding. Some forms of variation may be tolerated without substantially affecting communication, whereas others appear to be more strongly associated with intelligibility-related distinctions.

This interpretation is also compatible with Jenkins' Lingua Franca Core (Jenkins, 2000, 2002), which proposes that only a subset of pronunciation features consistently contribute to successful communication in English as a lingua franca. Rather than treating all pronunciation deviations as equally important, both cue-weighting approaches and the Lingua Franca Core suggest that intelligibility depends disproportionately on particular phonological contrasts.

The following sections discuss three phonological patterns that emerged particularly clearly in the present data: aspiration in bilabial plosives, the behavior of rhotics and laterals, and patterns related to vowels. These patterns illustrate how certain phonological dimensions may contribute more strongly to model disagreement and intelligibility assessment than others.

It must be noted that the instances of intelligible and unintelligible pronunciations are not distributed uniformly across the word list. As discussed in Section 3, the dataset contains slightly more intelligible than unintelligible recordings overall. However, this imbalance also varies considerably between individual words, as shown in Figure A1 in Appendix A. For example, words such as *paw*, *paws*, and *poor* contain a substantially higher proportion of unintelligible productions than words such as *boy*, *buy*, and *nose*.

This imbalance may influence the interpretation of some of the error analyses. Since certain phonological properties are concentrated in words that are intrinsically easier or more difficult, the observed effects may partly reflect characteristics of the word list rather than purely model-specific behavior. Consequently, while the examples and patterns discussed in the present chapter are representative of the dataset, they

should not necessarily be interpreted as evidence that the associated phonological features are universally more or less important for intelligibility. Instead, they should be understood in the context of the particular lexical and phonological composition of the word list used in the present study.

9.2.2 Bilabial Plosives

Aspiration errors were prominent, especially with voiceless bilabial plosives. The error analysis repeatedly identified words starting with */p/*as among the most problematic.

This pattern suggests that aspiration in bilabial plosives may provide comparatively weak acoustic information for the models to utilize. Unlike alveolar or velar stops, bilabial stops generally provide weaker and more diffuse acoustic place cues due to the absence of a front resonating cavity, resulting in less salient burst information and reduced spectral distinctiveness (Stevens and Blumstein, 1978; Green et al., 1997).

Consequently, the primary cue distinguishing aspirated and unaspirated realizations is often voice onset time (VOT), defined by Lisker and Abramson (1964) as the interval between release of a closure and the start of voicing. However, bilabial stops provide weaker and more diffuse acoustic place cues than alveolar or velar stops due to their reduced spectral distinctiveness (Stevens and Blumstein, 1978). Green et al. describe this as the "weak acoustic place cues associated with the /b/ auditory token" (Green et al., 1997), reflecting a broader characteristic of bilabials as a class. Because aspiration in bilabials relies more heavily on VOT in the absence of strong place cues, aspiration contrasts in words beginning with *p* may be particularly difficult to distinguish for both humans[1] and automatic classification systems.

The substitution analysis further supports this interpretation. Across all evaluated models, the most frequent substitution was the reduction of aspirated voiceless stops to their unaspirated counterparts, particularly /p^{h}/ → /p/. This substitution occurred substantially more often than any other individual error pattern and was observed consistently across models with otherwise very different architectures. The recurrence of this pattern suggests that aspiration represents a systematic challenge rather than an isolated weakness of a particular system.

The prominence of /p^{h}/ → /p/ substitutions also provides a likely explanation for the poor performance of many words beginning with /p/. In the word-level analyses, many of the words exhibiting the highest error rates began with aspirated bilabial stops. Taken together, these findings suggest that difficulties in representing aspiration, rather than broader problems with place or manner of articulation, may underlie a substantial proportion of the observed classification errors.

Interestingly, this finding highlights a tension between phonetic detail and intelligibility. Aspiration is included in the gold-standard transcriptions because Jenkins' Lingua Franca Core identifies it as a feature relevant for intelligibility in English (Jenkins, 2000, 2002). However, the feature-relaxation analysis showed that many aspiration-related mismatches occurred in productions that were nevertheless judged intelligible by human annotators. This suggests that while aspiration may contribute to pronunciation accuracy, its absence does not necessarily result in communication breakdown in all contexts.

[1] Interestingly, in our internship meetings, two research assistants who have been annotating noted that recognizing the aspiration in especially *p* was difficult.

One possible explanation is that aspiration occupies an intermediate position between phonetic detail and phonological contrast. Unlike voicing, which often directly distinguishes lexical items (e.g., *pat* vs. *bat*), aspiration in English is largely predictable from phonological context and is not typically used to distinguish words. Consequently, listeners may be relatively tolerant of variation in aspiration, particularly when other acoustic cues remain intact. In contrast, automatic speech-to-IPA systems must reproduce the phonetic form itself and therefore penalize aspiration mismatches more heavily than human listeners do during perception.

The prominence of aspiration errors illustrates one of the central findings of the present study. Although aspiration mismatches contributed substantially to transcription disagreement, many such productions remained intelligible to human listeners. This suggests that transcription-based evaluation may overestimate the communicative importance of certain phonetic deviations. The superior performance of the ensemble classifier likely stems in part from its ability to learn that aspiration mismatches do not necessarily imply communication breakdown. Rather than relying on individual transcription errors, the ensemble considers broader patterns of phonological variation when predicting intelligibility.

9.2.3 Rhotics and Laterals

The regression analyses revealed an interesting pattern for rhotics and laterals. In XGBoost, words containing rhotics and laterals were associated with increased error rates, particularly for false positives. Unlike aspiration, rhotics and laterals did not emerge as major sources of transcription disagreement in the feature-relaxation analysis. Nevertheless, they were strongly associated with classification errors in the regression analyses.

English rhotics are known to exhibit substantial acoustic variation across speakers, dialects, and phonetic environments (Ladefoged and Johnson, 2014). Unlike many consonant classes, rhotics do not have a single invariant acoustic realization and may be produced as approximants, taps, trills, or retroflex variants. This variability may increase the difficulty of automatically modeling rhotic segments, particularly when speakers have diverse linguistic backgrounds.

Similarly, laterals may exhibit considerable variation depending on syllable position. In English, dark and light variants of /l/ differ substantially in their acoustic realization, especially in word-final position. Consequently, words containing rhotics and laterals may present a more complex acoustic target than many other consonant classes.

The present findings, therefore, suggest that the elevated error rates associated with rhotics and laterals may reflect a combination of genuine phonetic variability and methodological factors. These features emerged as dominant sources of transcription disagreement, and they remained associated with classification errors in the regression analyses. This suggests that their influence on intelligibility assessment may be more indirect, arising from increased acoustic variability rather than from systematic phonological mismatches. Further work using segment-level analyses and manually verified alignments would be required to determine the precise source of these effects.

At the same time, caution is required when interpreting the apparent effects of consonantal features such as laterals. The segment alignment procedure is not unique, and different alignments can produce different assignments of substitutions, insertions, and deletions. In particular, liquids often occur in complex clusters and near vowel

transitions, making them sensitive to alignment decisions. Consequently, the elevated error rates associated with laterals should not necessarily be interpreted as evidence that /l/ itself is intrinsically difficult. Rather, they may partly reflect the interaction between phonetic variability and the assumptions imposed by the alignment algorithm. Vowel-related effects, which are less dependent on precise segment boundaries, are likely to be somewhat more robust.

9.2.4 Vowel-related patterns

In addition to consonantal effects, several statistically significant predictors identified in the regression analyses involved vowel properties. For IPA-Whisper, rounded vowels were associated with increased error rates, whereas diphthongs, back vowels, and long vowels were associated with lower error rates. Similarly, long vowels reduced false-negative rates for XGBoost.

Unlike aspiration and liquids, these effects did not correspond to particularly prominent patterns in the word-level analyses. Instead, they appeared to be distributed across many words, suggesting that vowel-related influences are more diffuse and less easily attributed to a small number of problematic lexical items.

The interpretation of these effects is not straightforward. Rounded vowels may be particularly susceptible to variation in non-native speech, whereas diphthongs and long vowels may provide additional acoustic information that facilitates recognition. However, since the present analyses were conducted at the word level rather than the segment level, the precise source of these effects remains unclear. Consequently, these findings should be interpreted with caution and would benefit from further investigation through segment-level analyses and manually verified alignments.

9.3 Checking for lexicality bias

As discussed in Chapter 2, some speech-to-IPA models may exhibit lexical bias. To investigate this, a post-hoc analysis was conducted to assess whether lexical frequency influenced model performance. Word frequencies were obtained using Zipf frequency scores using the *wordfreq* database (Speer, 2022). Correlations between lexical frequency and word-level error rates were examined for both IPA-Whisper and XGBoost.

Lexicality in the word list

The word list is skewed toward common words. The average Zipf frequency is 4.41 (σ = 0.70), with most of the words falling into the *common* (28 words) or *very common* (12 words) categories. Only a single word, *seedy*, falls into the *rare* category. The lexicality frequency is shown in Table 9.1.

Because the frequency bands were unevenly distributed and showed substantial overlap, they were not interpreted as evidence for a systematic frequency effect. Instead, the scatterplots and correlation coefficients were used as the main post-hoc lexicality analysis. These showed no meaningful relationship between Zipf frequency and overall error rate for either model.

Table 9.1: Zipf frequency bands used in the lexicality analysis.

Frequency band	Zipf frequency range	n words	Example word(s)
Rare	Zipf < 3	1	*seedy*
Uncommon	$3 \leq$ Zipf < 4	11	*putt*, *farther*, *paw*
Common	$4 \leq$ Zipf < 5	28	*tide*, *dance*, *fair*
Very common	Zipf ≥ 5	12	*father*, *boy*, *city*

IPA-Whisper

The correlation between lexical frequency and overall error rate is essentially zero (ρ = -0.02), and the regression line in the scatterplot is almost perfectly horizontal (see Figure C1 in Appendix C)

The strongest association is between lexical frequency and FP rates (ρ = 0.36) (see Figure C3 in Appendix C), indicating that more common words are somewhat more likely to be incorrectly accepted as intelligible. It must be noted that IPA-Whisper tends to overpredict the unintelligible class, so this relationship should be interpreted with caution.

FN rates show only a weak negative relationship with frequency (ρ = -0.12) (see Figure C2 in Appendix C), suggesting that more frequent words are only slightly less likely to be incorrectly rejected.

XGBoost

The correlation between lexical frequency and error rates of XGBoost is near zero (ρ = -0.07), and the regression line is almost flat (see Figure C4 in Appendix C). This indicates that common words are not systematically easier to classify than less common words.

There is a weak tendency for more frequent words to have lower FN rates (ρ = -0.17), meaning that common words are slightly less likely to be incorrectly classified as unintelligible (see Figure C5 in Appendix C). Conversely, more frequent words show slightly higher FP rates (ρ = -0.17) (see Figure C6 in Appendix C). However, both effects are small and unlikely to be practically important.

Summary and Interpretation

Overall, the lexicality analysis provides little evidence that lexical frequency substantially influenced model performance. Correlations between lexical frequency and error rates were weak for both IPA-Whisper and XGBoost, and the lexical feature ablation experiment produced virtually no change in classification performance. Together, these findings suggest that the ensemble classifier relies primarily on pronunciation-related information rather than lexical familiarity. However, because the word list contains relatively few low-frequency items, this conclusion should be interpreted cautiously.

9.4 Evaluation of the hypotheses

The first hypothesis predicted that models operating at a phonetic level would outperform phoneme-based systems. This expectation was only partially supported. Although

IPA-Whisper achieved the strongest overall performance among the individual models, the two Wav2Vec2Phoneme systems performed competitively and substantially outperformed both MultIPA and Allosaurus. These findings suggest that access to detailed phonetic representations alone does not guarantee improved performance. Factors such as training data, language coverage, and acoustic modeling quality may be equally important.

The second hypothesis predicted that multilingual phonetic models would better capture non-native variation because they are exposed to diverse speech data. This hypothesis was not supported. Contrary to expectations, Allosaurus and MultIPA produced the weakest overall results. One possible explanation is that multilingual training promotes broad phonetic coverage at the expense of language-specific precision. Since the present task involved English-learner speech, models with stronger English representations may have been better suited to it.

The third hypothesis proposed a trade-off between stability and phonetic detail. The results provide partial support for this prediction. The phoneme-based systems often produced more stable outputs, while the phonetic systems were capable of representing additional detail such as aspiration and vowel variation. However, this additional detail did not necessarily translate into improved performance in intelligibility classification.

Finally, the second research question predicted that combining multiple systems in an ensemble model would improve classification performance. This hypothesis was strongly supported. The XGBoost classifier substantially outperformed all individual speech-to-IPA systems, demonstrating that the strengths of different models can be combined to produce a more accurate representation of intelligibility.

9.5 Answering the research question

This study addressed two research questions. The first asked how effectively off-the-shelf speech-to-IPA models can be used for intelligibility classification. The second investigated whether the strengths of multiple speech-to-IPA systems could be combined to improve classification performance.

Regarding the first research question, the results indicate that speech-to-IPA models provide useful information for intelligibility assessment but are most likely not sufficiently accurate as standalone intelligibility classifiers. Although IPA-Whisper achieved the strongest overall performance among the individual models, all evaluated systems exhibited substantial limitations when transcription agreement was used as a proxy for intelligibility. The analyses revealed systematic phonological error patterns, particularly involving aspiration, voicing, and vowel length. They demonstrated that many transcription mismatches occurred in productions that were nevertheless judged intelligible by human annotators. These findings suggest that intelligibility cannot be reduced to transcription accuracy alone.

The second research question asked whether the strengths of multiple speech-to-IPA systems could be combined to improve classification performance. The results provide a clear positive answer. The XGBoost ensemble classifier substantially outperformed all individual speech-to-IPA models and achieved more balanced performance across both intelligible and unintelligible classes. Furthermore, the ensemble also outperformed the intelligibility classification systems developed in previous internship studies by Zou (2024) and Kuan (2025). This improvement suggests that the different speech-to-IPA models capture complementary aspects of pronunciation and that combining

information across systems provides a more robust representation of intelligibility than any individual model alone.

Taken together, the findings suggest that intelligibility assessment is best understood as a problem of cue integration rather than direct transcription matching. Individual speech-to-IPA systems capture only part of the phonological information relevant to listener understanding. In contrast, ensemble approaches can combine multiple sources of evidence into a more comprehensive representation. The results, therefore, demonstrate that speech-to-IPA systems constitute a valuable source of information for automatic intelligibility assessment, but that their greatest utility may lie in serving as components of larger assessment frameworks rather than as standalone classifiers.

More broadly, the findings support the distinction between pronunciation accuracy and intelligibility established in previous research (Munro and Derwing, 1999). While accurate phonetic representations remain important, successful intelligibility assessment requires consideration of which phonological deviations are likely to affect listener understanding. The present results suggest that this information is captured more effectively through patterns of phonological variation and cue integration than through direct comparison with canonical transcriptions alone.

9.6 Limitations

Several limitations should be considered when interpreting the findings of the present study.

First, intelligibility is approximated indirectly through transcription-based representations. Speech-to-IPA systems are designed to reproduce phonetic structure rather than model human speech perception. Consequently, exact transcription mismatches do not necessarily correspond to communication breakdown. Although the ensemble model partially alleviates this problem by learning which patterns of mismatch are associated with unintelligibility, the underlying representation remains transcription-centered rather than perception-centered.

Second, the gold-standard IPA transcriptions represent an abstraction rather than an objective ground truth. Even with multiple accepted variants, the reference transcriptions inevitably simplify natural phonetic variation. Some apparent model errors may therefore reflect limitations of the gold standard rather than genuine pronunciation problems.

Third, the speech-to-IPA systems themselves are imperfect and may introduce systematic biases. The models were trained on different datasets, employ different architectures, and operate at different representational levels. Consequently, some of the phonological patterns identified in the present study may reflect model-specific tendencies rather than properties of learner speech. While using multiple models and an ensemble approach reduces dependence on any single system, model-specific biases cannot be entirely ruled out.

Fourth, the alignment procedures used throughout the study impose assumptions about segment correspondence. Insertions, deletions, and substitutions are determined algorithmically, and different alignment strategies could produce somewhat different error counts and feature-level patterns. The reported segmental errors should therefore be interpreted as approximations rather than unique solutions.

Fifth, although a qualitative inspection of model outputs was conducted on a subset of the recordings, the present study did not include a systematic manual phonetic

analysis of all identified error patterns. Consequently, some interpretations remain inferential and would benefit from targeted acoustic or perceptual validation.

Sixth, the analyses also rely on a multi-stage processing pipeline in which later stages depend on earlier preprocessing decisions. Feature-level comparisons, substitution analyses, and ensemble features all ultimately derive from the normalized IPA transcriptions and alignment procedure. Consequently, biases or errors introduced at earlier stages may propagate through subsequent analyses.

Seventh, the present study evaluates a single prediction from each speech-to-IPA system for every recording. Although this approach reflects typical usage, it does not assess the stability of model outputs across different decoding settings, software versions, or repeated runs. Future work could investigate the robustness of the observed patterns under alternative decoding strategies and prediction settings.

Finally, the present study adopts a relatively narrow operationalization of intelligibility. The data consist of isolated words rather than connected speech, and intelligibility is represented as a binary distinction. In natural communication, intelligibility is influenced by context, prosody, coarticulation, listener expectations, and discourse-level information. The findings should therefore be interpreted as applying primarily to word-level intelligibility rather than to intelligibility in spontaneous speech.

9.7 Further Research

The present study demonstrates that intelligibility can be predicted more accurately by combining information from multiple speech representations than by relying on direct IPA comparison alone. Several directions for future research emerge from these findings.

One possible direction is to develop an interpretable weighted IPA comparison framework. Rather than treating all transcription mismatches equally, such a system would assign different weights to different phonological features according to their relative contribution to intelligibility. The results of the present study suggest, for example, that aspiration mismatches often have relatively little impact on intelligibility, whereas other phonological properties appear to be more informative. In many respects, this is the type of weighting that the ensemble classifier appears to learn automatically from the feature-level error counts. However, an explicit weighting framework would make these decisions transparent and directly interpretable. Such a system could therefore occupy an intermediate position between strict transcription matching and machine-learning-based intelligibility prediction, combining linguistic interpretability with improved sensitivity to perceptually relevant pronunciation differences.

A second direction concerns the diversity of information available to the ensemble classifier. The present study combined five speech-to-IPA systems that produce relatively similar phonetic or phonological representations. Future work could investigate whether incorporating more heterogeneous systems further improves performance. For example, conventional automatic speech recognition (ASR) systems could contribute lexical information, while pronunciation-scoring models, forced-alignment systems, or acoustic confidence measures could provide complementary evidence regarding pronunciation quality. Similarly, using n-best hypotheses from ASR or speech-to-IPA systems, rather than relying solely on the single best prediction, may better capture pronunciation ambiguity and reduce the impact of individual transcription errors.

Another promising direction stems from the ongoing development of a more detailed

manual annotation layer for the MOOC dataset. In addition to binary intelligibility judgments, this annotation records specific pronunciation phenomena, such as aspiration, vowel quality, vowel length, rhoticity, consonant substitutions, and missing or extra sounds. Such annotations would enable training models to predict not only whether a pronunciation is intelligible but also which pronunciation errors are present. This could support more fine-grained pronunciation feedback, enable direct evaluation of speech-to-IPA systems at the error-type level, and provide richer supervision for future pronunciation assessment models.

Finally, future work could expand the phonological representations used in the present study. The simplified feature inventory was designed to maximize interpretability and reduce computational complexity, but additional articulatory, acoustic, or prosodic features may provide further insight into pronunciation errors. Likewise, the gold-standard IPA inventory could be expanded to include a broader range of acceptable pronunciation variants or weighted by their perceptual acceptability. Such refinements may improve both transcription-based evaluation and downstream intelligibility prediction.

Chapter 10

Conclusion

This thesis investigated whether off-the-shelf speech-to-IPA models can be used to classify the intelligibility of L2 English single-word pronunciations and whether the strengths of multiple systems can be combined to improve classification performance.

Four types of speech-to-IPA models were evaluated using recordings collected from learners participating in the MOOC *English Pronunciation in a Global World*. The models were first compared against gold-standard IPA transcriptions using direct transcription matching and feature-based analyses. The resulting phonological information was then used to construct ensemble classifiers for intelligibility prediction.

The results showed that individual speech-to-IPA systems provide useful information about pronunciation quality but are not sufficiently accurate as standalone intelligibility classifiers. Among the evaluated systems, IPA-Whisper achieved the strongest overall performance. However, all models exhibited substantial limitations when transcription agreement was used as a direct proxy for intelligibility. Feature-based analyses revealed systematic error patterns across models, with aspiration emerging as one of the most prominent sources of mismatch. Importantly, many of these deviations occurred in productions that human annotators nevertheless judged intelligible, highlighting the distinction between phonetic accuracy and listener understanding.

The ensemble approach proved considerably more successful. By combining phonological error patterns, feature-level mismatches, and agreement information across multiple speech-to-IPA systems, the XGBoost classifier substantially outperformed all individual models. These findings demonstrate that intelligibility is not captured solely by transcription accuracy but rather emerges from broader patterns of phonological variation. The results, therefore, suggest that speech-to-IPA systems are most effective when used as sources of phonological evidence within larger classification frameworks rather than as standalone intelligibility predictors.

More broadly, the findings support the view that intelligibility assessment is best understood as a problem of cue integration rather than simple pronunciation matching. Different phonological features contribute unequally to intelligibility, and not all deviations from canonical pronunciations result in communication breakdown. Consequently, future pronunciation assessment systems may benefit from moving beyond strict transcription comparison and instead modeling the relative contribution of different phonological cues to listener understanding.

Overall, this study demonstrates that speech-to-IPA models provide a valuable foundation for automatic intelligibility assessment. While substantial challenges remain, particularly regarding the relationship between phonetic variation and listener

perception, the results show that combining phonological information from multiple systems can produce meaningful improvements in intelligibility classification and offer a promising direction for future pronunciation assessment research.

Appendix

Appendix A: Figures & Tables

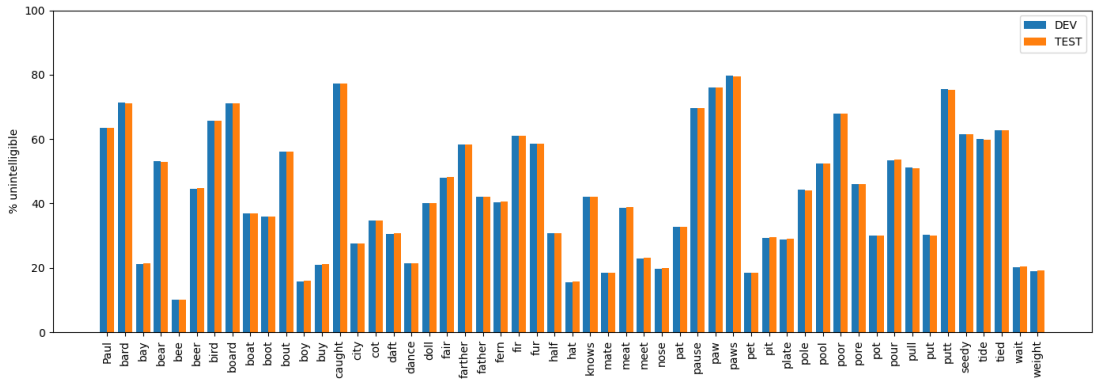


Figure A1: Proportion of unintelligible productions by word in the development and test sets

Table A3: Descriptions of ensemble features.

Feature	Description
`word`	Target word identity.
`binary`	Binary model output, where 1 indicates that the model prediction matched an acceptable gold pronunciation and 0 indicates that it did not.
`n_gold_segments`	Number of segments in the selected gold-standard pronunciation.
`n_pred_segments`	Number of segments in the model prediction.
`length_diff`	Difference between the number of predicted and gold segments.
`segment_errors`	Number of segment-level mismatches, insertions, and deletions.
`missing`	Number of gold segments missing from the model prediction.
`extra`	Number of additional segments predicted by the model.
`consonant_errors`	Number of consonant-related mismatches.
`vowel_errors`	Number of vowel-related mismatches.
`aspiration_errors`	Number of aspiration mismatches.
`voicing_errors`	Number of voicing mismatches.

Feature	Description
`place_errors`	Number of place-of-articulation mismatches.
`manner_errors`	Number of manner-of-articulation mismatches.
`height_errors`	Number of vowel-height mismatches.
`backness_errors`	Number of vowel-backness mismatches.
`round_errors`	Number of lip-rounding mismatches.
`long_errors`	Number of vowel-length mismatches.
`diph_errors`	Number of diphthong mismatches.
`*_rate`	Normalized version of the corresponding error count, calculated relative to the relevant segment or comparison count.
`pred_gold_ratio`	Ratio between the number of predicted and gold segments.
`total_feature_errors`	Total number of feature-level mismatches.
`consonant_minus_vowel_errors`	Difference between consonant and vowel error counts.
`vote_sum`	Sum of binary predictions across the five models.
`vote_mean`	Mean binary prediction across the five models.
`vote_std`	Standard deviation of binary predictions across the five models.
`all_models_agree`	Binary indicator showing whether all models made the same prediction.
`n_models_positive`	Number of models predicting the positive class.
`n_models_negative`	Number of models predicting the negative class.
`vote_entropy`	Entropy of the model vote distribution.
`agree_ModelA_ModelB`	Binary indicator showing whether two specific models made the same prediction.

Figure A2: Example of feature bundle

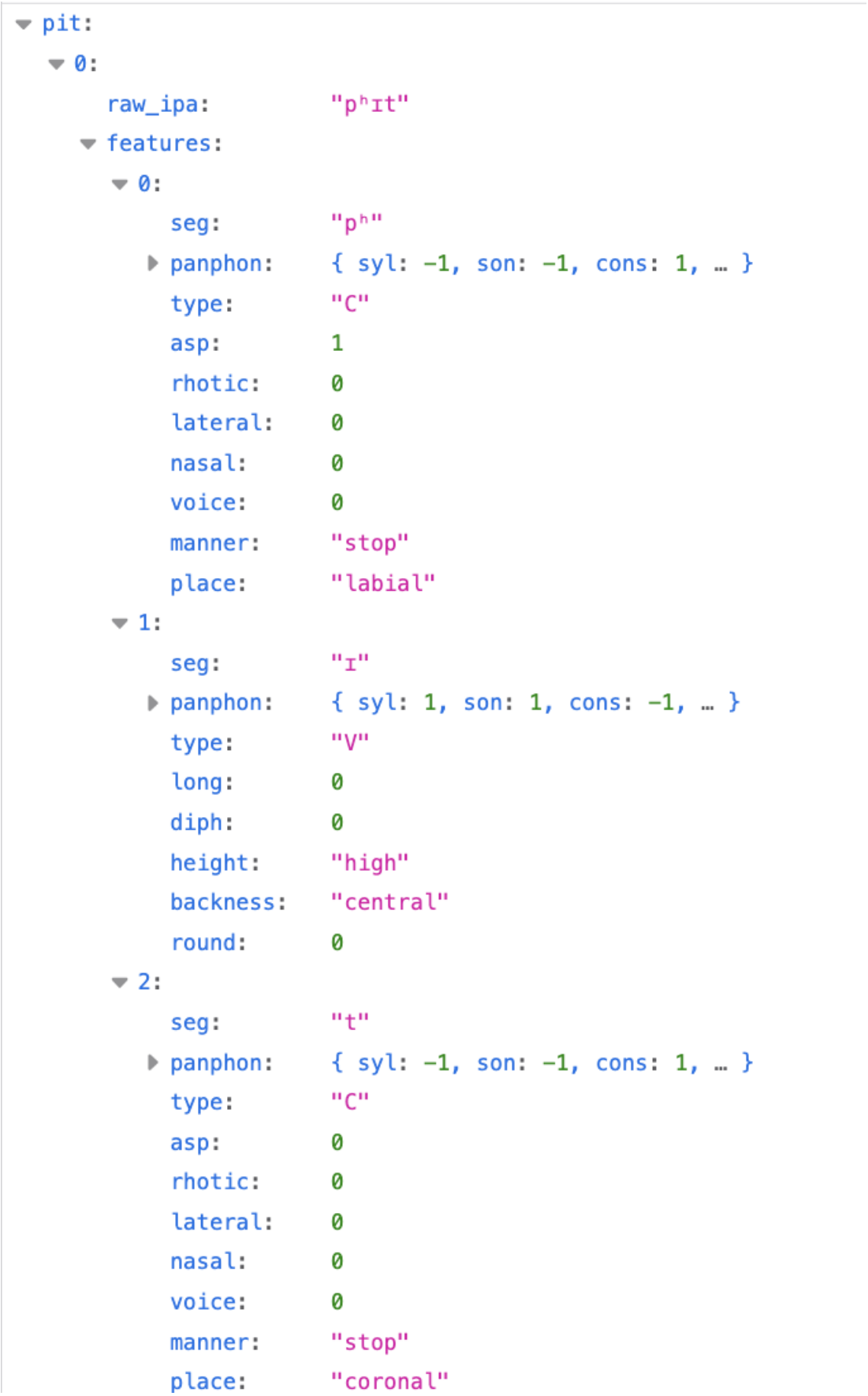


Table A1: Example IPA-to-feature mapping for consonants

IPA	place	manner	voice	asp	nasal	lateral	rhotic
p	bilabial	stop	0	0	0	0	0
pʰ	bilabial	stop	0	1	0	0	0
b	bilabial	stop	1	0	0	0	0
t	alveolar	stop	0	0	0	0	0
d	alveolar	stop	1	0	0	0	0
k	velar	stop	0	0	0	0	0
g	velar	stop	1	0	0	0	0
m	bilabial	nasal	1	0	1	0	0
n	alveolar	nasal	1	0	1	0	0
ŋ	velar	nasal	1	0	1	0	0
s	alveolar	fricative	0	0	0	0	0
z	alveolar	fricative	1	0	0	0	0
l	alveolar	approximant	1	0	0	1	0
r	alveolar	trill	1	0	0	0	1
j	palatal	approximant	1	0	0	0	0

Table A2: Example IPA-to-feature mapping for vowels

IPA	height	backness	round	long	diph
i	high	front	0	0	0
iː	high	front	0	1	0
y	high	front	1	0	0
u	high	back	1	0	0
e	mid	front	0	0	0
ø	mid	front	1	0	0
o	mid	back	1	0	0
a	low	central	0	0	0
ai	low-front	front	0	0	1
ou	mid-back	back	1	0	1

Table A4: Feature relaxation analysis: accuracy gains after ignoring individual phonological features by intelligibility label.

		Unintelligible				Intelligible			
Model	Feature	Orig.	Relaxed	Gain	Errors	Orig.	Relaxed	Gain	Errors
XLSR-53	asp	0.102	0.206	+0.104	0.362	0.328	0.464	+0.136	0.241
	voice	0.102	0.125	+0.023	0.103	0.328	0.353	+0.025	0.059
	long	0.102	0.119	+0.017	0.049	0.328	0.361	+0.033	0.054
	place	0.102	0.107	+0.005	0.040	0.328	0.347	+0.019	0.052
	manner	0.102	0.112	+0.010	0.040	0.328	0.346	+0.018	0.031
	height	0.102	0.112	+0.010	0.064	0.328	0.330	+0.002	0.030
	backness	0.102	0.102	+0.000	0.013	0.328	0.328	+0.000	0.008
	round	0.102	0.103	+0.001	0.005	0.328	0.329	+0.001	0.002
	diph	0.102	0.102	+0.000	0.000	0.328	0.328	+0.000	0.000
	lateral	0.102	0.102	+0.000	0.000	0.328	0.328	+0.000	0.000
	nasal	0.102	0.102	+0.000	0.000	0.328	0.328	+0.000	0.000
	rhotic	0.102	0.102	+0.000	0.000	0.328	0.328	+0.000	0.000
IPA-Whisper	asp	0.094	0.172	+0.078	0.351	0.346	0.472	+0.126	0.237
	voice	0.094	0.141	+0.047	0.129	0.346	0.383	+0.037	0.063
	long	0.094	0.117	+0.023	0.049	0.346	0.393	+0.047	0.071
	place	0.094	0.097	+0.003	0.021	0.346	0.361	+0.015	0.031
	manner	0.094	0.102	+0.008	0.028	0.346	0.357	+0.012	0.020
	height	0.094	0.101	+0.007	0.060	0.346	0.352	+0.006	0.028
	backness	0.094	0.101	+0.007	0.020	0.346	0.348	+0.003	0.007
	round	0.094	0.094	+0.000	0.001	0.346	0.346	+0.000	0.004
	diph	0.094	0.094	+0.000	0.000	0.346	0.346	+0.000	0.000
	lateral	0.094	0.094	+0.000	0.000	0.346	0.346	+0.000	0.000
	nasal	0.094	0.094	+0.000	0.000	0.346	0.346	+0.000	0.000
	rhotic	0.094	0.094	+0.000	0.000	0.346	0.346	+0.000	0.000
LV-60	asp	0.093	0.159	+0.066	0.369	0.312	0.426	+0.114	0.243
	voice	0.093	0.121	+0.029	0.135	0.312	0.353	+0.041	0.101
	long	0.093	0.115	+0.022	0.071	0.312	0.359	+0.047	0.069
	place	0.093	0.096	+0.003	0.040	0.312	0.321	+0.009	0.051
	manner	0.093	0.104	+0.012	0.045	0.312	0.326	+0.014	0.027
	height	0.093	0.103	+0.010	0.060	0.312	0.317	+0.004	0.048
	backness	0.093	0.093	+0.000	0.012	0.312	0.312	+0.000	0.004
	round	0.093	0.097	+0.005	0.008	0.312	0.313	+0.001	0.001
	diph	0.093	0.093	+0.000	0.000	0.312	0.312	+0.000	0.000
	lateral	0.093	0.093	+0.000	0.000	0.312	0.312	+0.000	0.000
	nasal	0.093	0.093	+0.000	0.000	0.312	0.312	+0.000	0.000
	rhotic	0.093	0.093	+0.000	0.000	0.312	0.312	+0.000	0.000
MultiIPA	asp	0.045	0.120	+0.076	0.376	0.117	0.230	+0.113	0.280
	voice	0.045	0.084	+0.039	0.190	0.117	0.153	+0.036	0.163
	long	0.045	0.061	+0.016	0.055	0.117	0.142	+0.025	0.060
	place	0.045	0.050	+0.006	0.062	0.117	0.141	+0.024	0.082
	manner	0.045	0.049	+0.005	0.054	0.117	0.131	+0.014	0.048
	height	0.045	0.049	+0.005	0.054	0.117	0.124	+0.007	0.051
	backness	0.045	0.046	+0.001	0.031	0.117	0.120	+0.003	0.020
	round	0.045	0.045	+0.000	0.001	0.117	0.117	+0.000	0.002
	diph	0.045	0.045	+0.000	0.000	0.117	0.117	+0.000	0.000
	lateral	0.045	0.045	+0.000	0.000	0.117	0.117	+0.000	0.000
	nasal	0.045	0.045	+0.000	0.000	0.117	0.117	+0.000	0.000
	rhotic	0.045	0.045	+0.000	0.000	0.117	0.117	+0.000	0.000
Allosaurus	asp	0.022	0.028	+0.006	0.093	0.046	0.054	+0.008	0.072
	voice	0.022	0.029	+0.007	0.062	0.046	0.054	+0.008	0.054
	place	0.022	0.028	+0.006	0.127	0.046	0.059	+0.013	0.133
	long	0.022	0.025	+0.003	0.026	0.046	0.059	+0.012	0.037
	manner	0.022	0.023	+0.001	0.030	0.046	0.050	+0.004	0.038
	height	0.022	0.025	+0.003	0.102	0.046	0.053	+0.007	0.070
	backness	0.022	0.022	+0.000	0.003	0.046	0.048	+0.002	0.008
	round	0.022	0.023	+0.001	0.010	0.046	0.046	+0.000	0.011
	diph	0.022	0.022	+0.000	0.000	0.046	0.046	+0.000	0.000
	lateral	0.022	0.022	+0.000	0.000	0.046	0.046	+0.000	0.000
	nasal	0.022	0.022	+0.000	0.000	0.046	0.046	+0.000	0.000
	rhotic	0.022	0.022	+0.000	0.000	0.046	0.046	+0.000	0.000

Table A5: Feature-level error-rates by model.

feature	Allosaurus	IPA-Whisper	MultiIPA	W2V2-LV60	W2V2-XLSR
asp	0.454	0.292	0.307	0.329	0.327
backness	0.306	0.122	0.180	0.131	0.141
diph	0.515	0.273	0.356	0.279	0.287
height	0.534	0.254	0.312	0.275	0.265
lateral	0.337	0.136	0.150	0.154	0.158
long	0.320	0.286	0.338	0.322	0.348
manner	0.524	0.203	0.259	0.234	0.232
nasal	0.358	0.131	0.145	0.165	0.162
place	0.561	0.173	0.270	0.202	0.203
rhotic	0.344	0.148	0.172	0.184	0.177
round	0.448	0.176	0.220	0.168	0.173
type	0.278	0.103	0.114	0.120	0.124
voice	0.454	0.201	0.270	0.254	0.233

Table A6: Final optimized hyperparameters for the XGBoost model trained on the complete DEV set prior to TEST evaluation.

Parameter	Value
`n_estimators`	250
`max_depth`	2
`learning_rate`	0.03
`subsample`	1.0
`colsample_bytree`	0.7

Appendix B : GOLD IPA

Table B1: Target words and corresponding GOLD IPA transcriptions used in the study.

Word	IPA Transcriptions
pit	/pʰɪt/
pet	/pʰet/, /pʰɛt/
pat	/pʰat/, /pʰæt/, /pʰɐt/
put	/pʰut/, /pʰʊt/
putt	/pʰʌt/
pot	/pʰot/, /pʰɑt/, /pʰɒt/, /pʰɔt/
bee	/biː/, /bɪː/
bay	/bei/, /beɪ/, /bɛi/, /bɛɪ/
buy	/bai/, /baɪ/, /bʌi/, /bʌɪ/
boy	/boi/, /boɪ/, /bɔi/, /bɔɪ/
boot	/but/, /buːt/, /bɯt/, /bɯːt/, /bʊːt/, /bʊt/
boat	/bout/, /boʊt/, /bəut/, /bəʊt/
bout	/baut/, /baʊt/
beer	/bɪəɹ/, /bɪɹ/, /bɪːɹ/
bear	/beəɹ/, /beɹ/, /beːɹ/, /bɛə/, /bɛəɹ/, /bɛɹ/, /bɛːɹ/
bird	/beːɹd/, /bəːɹd/, /bɚːd/, /bɜːɹd/, /bɜ˞ːd/, /beɹd/, /bəɹd/, /bɚd/, /bɜɹd/, /bɜ˞d/
bard	/baɹd/, /bɑːɹd/, /bɑɹd/, /baːɹd/
board	/boɹd/, /bɒd/, /bɒɹd/, /bɔːd/, /bɔːɹd/
city	/siti/, /sɪti/
seedy	/siːdi/, /sɪːdi/
hat	/hat/, /hæt/
dance	/dans/, /dæns/, /dɑːns/, /dæːns/
daft	/daft/, /dæft/, /dɑːft/, /dɑft/, /daːft/, /dæːft/
half	/haf/, /hæf/, /hɑːf/, /hæːf/, /hɑf/, /haːf/
father	/faðə/, /faðɚ/, /fɐðə/, /fɐðɚ/, /fɑːðə/, /fɑːðɚ/
farther	/faɹðɚ/, /fɐɹðɚ/, /fɑːɹðɚ/
pull	/pʰul/, /pʰʊl/, /pʰuɬ/, /pʰʊɬ/
pool	/pʰuːl/, /pʰuːɬ/
pole	/pʰoul/, /pʰoʊl/, /pʰəul/, /pʰəʊl/, /pʰouɬ/, /pʰoʊɬ/, /pʰəuɬ/, /pʰəʊɬ/
paul	/pʰɔl/, /pʰɔːl/, /pʰɔɬ/, /pʰɔːɬ/
doll	/dol/, /dɑl/, /dɒl/, /dɔl/, /doɬ/, /dɑɬ/, /dɒɬ/, /dɔɬ/

Word	IPA Transcriptions
cot	/kʰot/, /kʰɑt/, /kʰɒt/
caught	/kʰɑːt/, /kʰɔːt/
fir	/fɚ/, /fɚː/, /fɜɹ/, /fɜːɹ/, /fɝ/, /fɝː/
fern	/fɚn/, /fɚːn/, /fɜɹn/, /fɜːɹn/, /fɝn/, /fɝːn/
fur	/fɚ/, /fɜɹ/, /fɜːɹ/, /fɝ/
fair	/feəɹ/, /feɹ/, /fɛəɹ/, /fɛɹ/
nose	/nouz/, /noʊz/, /nəuz/, /nəʊz/
knows	/nouz/, /noʊz/, /nəuz/, /nəʊz/
plate	/pʰleit/, /pʰleɪt/, /pʰlɛit/, /pʰlɛɪt/, /pleit/, /pleɪt/, /plɛit/, /plɛɪt/
weight	/weit/, /weɪt/, /wɛit/, /wɛɪt/
poor	/pʰɔːɹ/, /pʰʊːɹ/
pour	/pʰɒːɹ/, /pʰɒɹ/, /pʰɔːəɹ/, /pʰɔːɹ/
pore	/pʰɒːɹ/, /pʰɔːɹ/
paw	/pʰɑː/, /pʰɒ/, /pʰɔː/
tide	/tʰaɪd/, /tʰɐɪd/
tied	/tʰaɪd/, /tʰɐɪd/
pause	/pʰɑːz/, /pʰɒz/, /pʰɔːz/, /pʰɔz/, /pʰɒːz/
paws	/pʰɑːz/, /pʰɔːz/, /pʰɒːz/
meet	/miːt/
meat	/miːt/, /miːt/
mate	/meit/, /meɪt/, /mɛit/, /mɛɪt/

Appendix C: Post-hoc

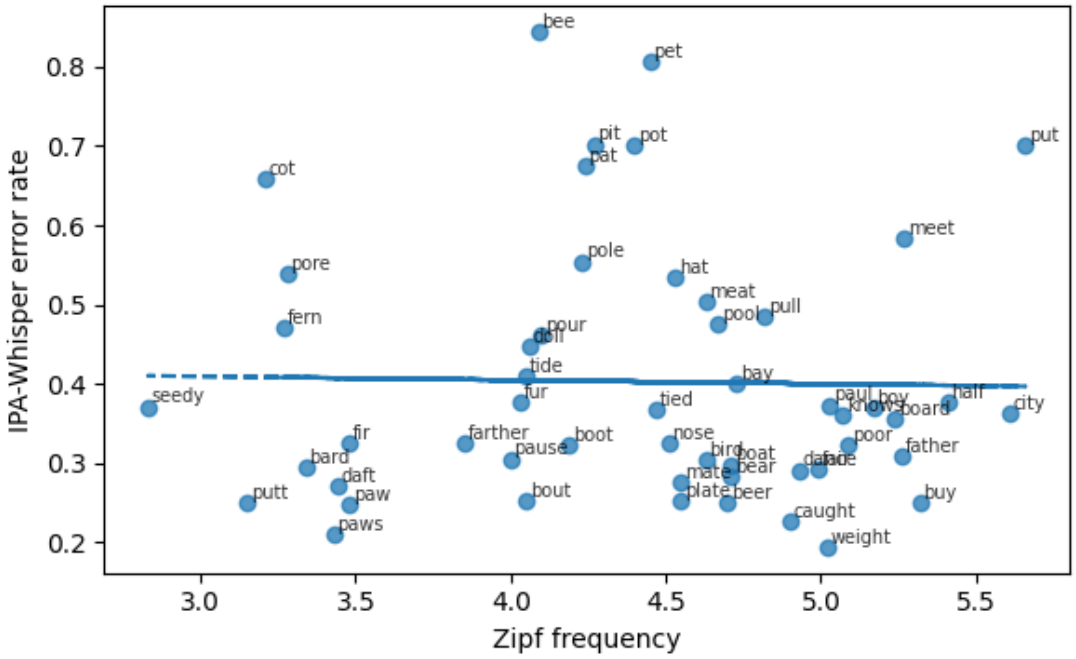


Figure C1: IPA-Whisper error rate by lexical frequency

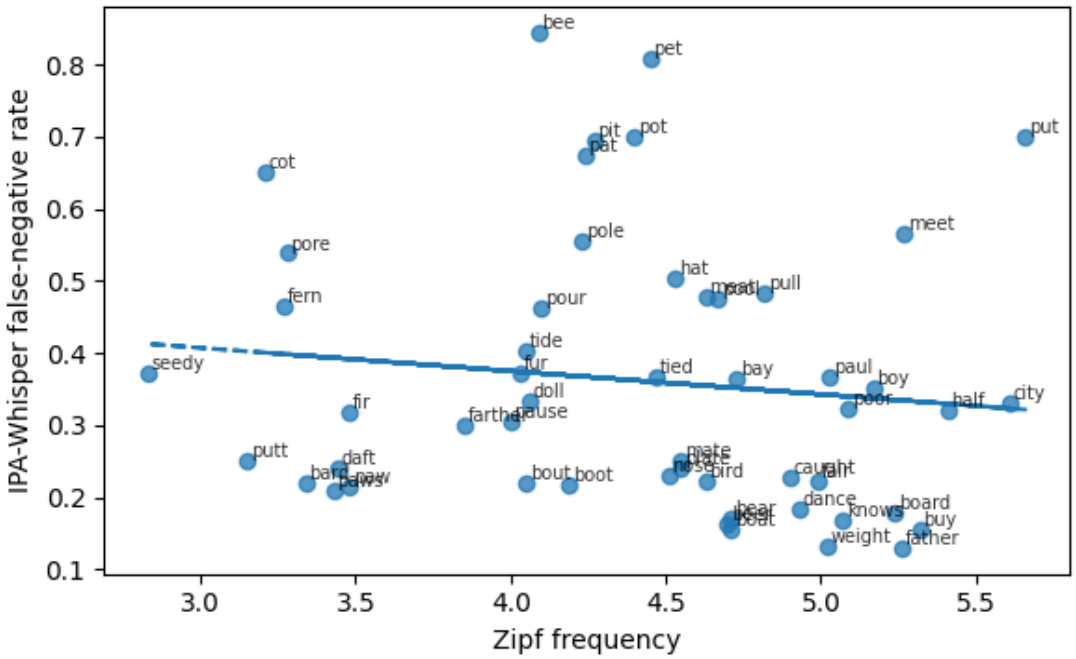


Figure C2: IPA-Whisper FN rate by lexical frequency

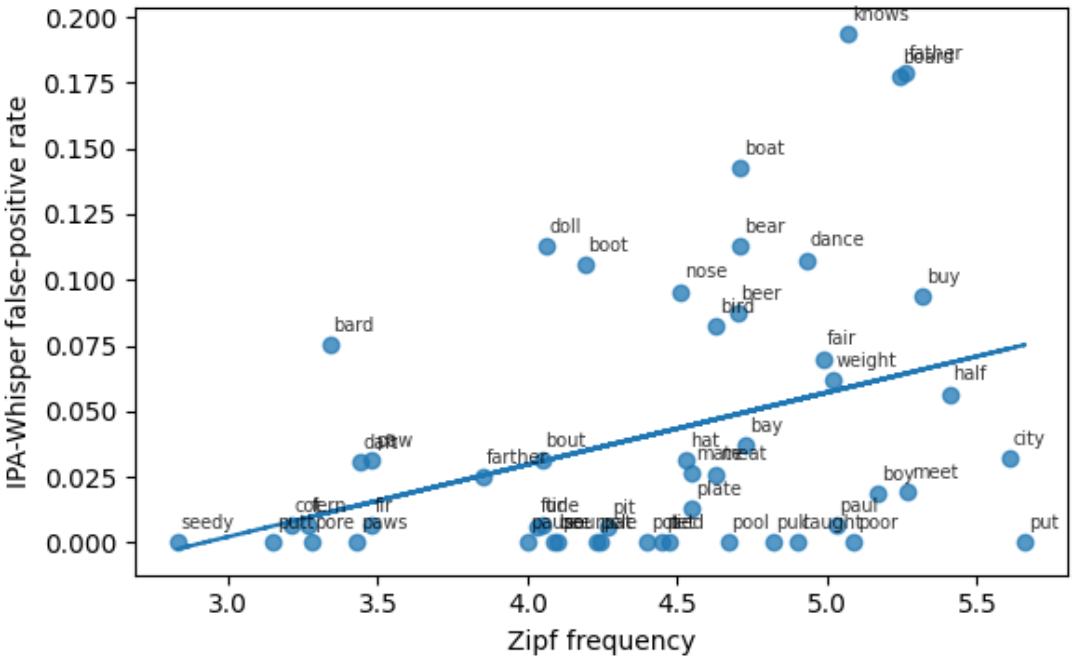


Figure C3: IPA-Whisper FP rate by lexical frequency

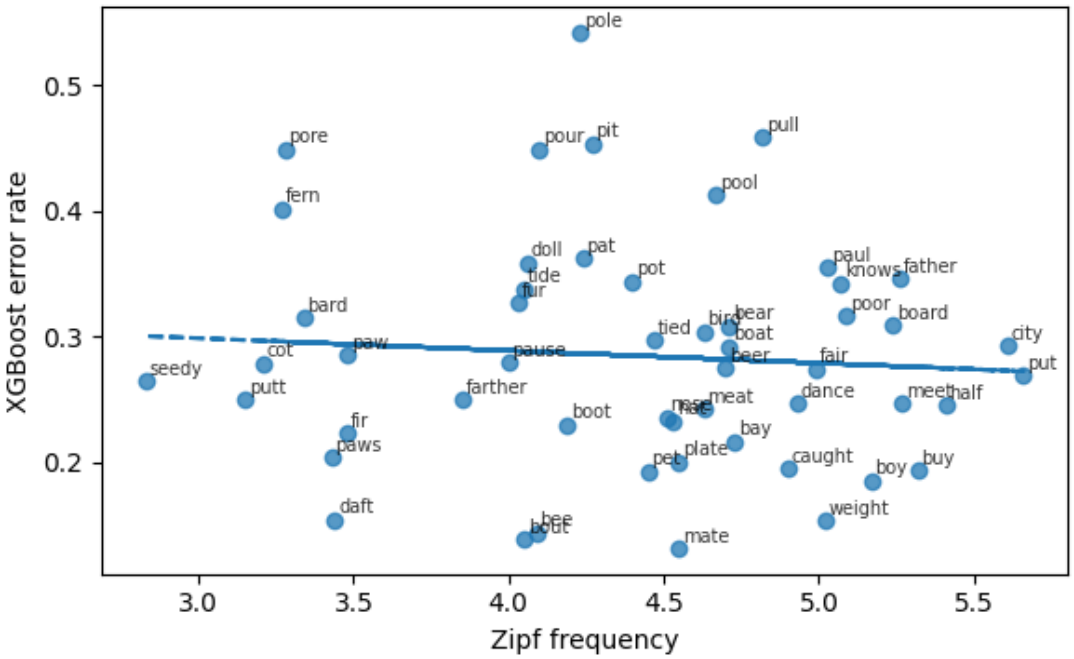


Figure C4: XGBoost error rate by lexical frequency

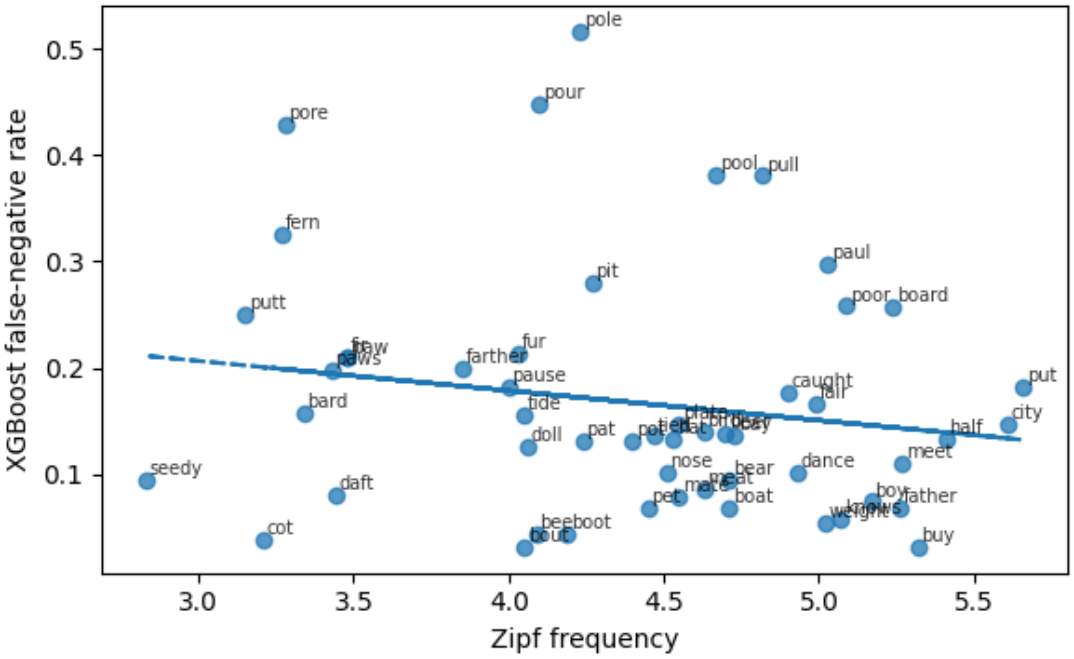


Figure C5: XGBoost FN rate by lexical frequency

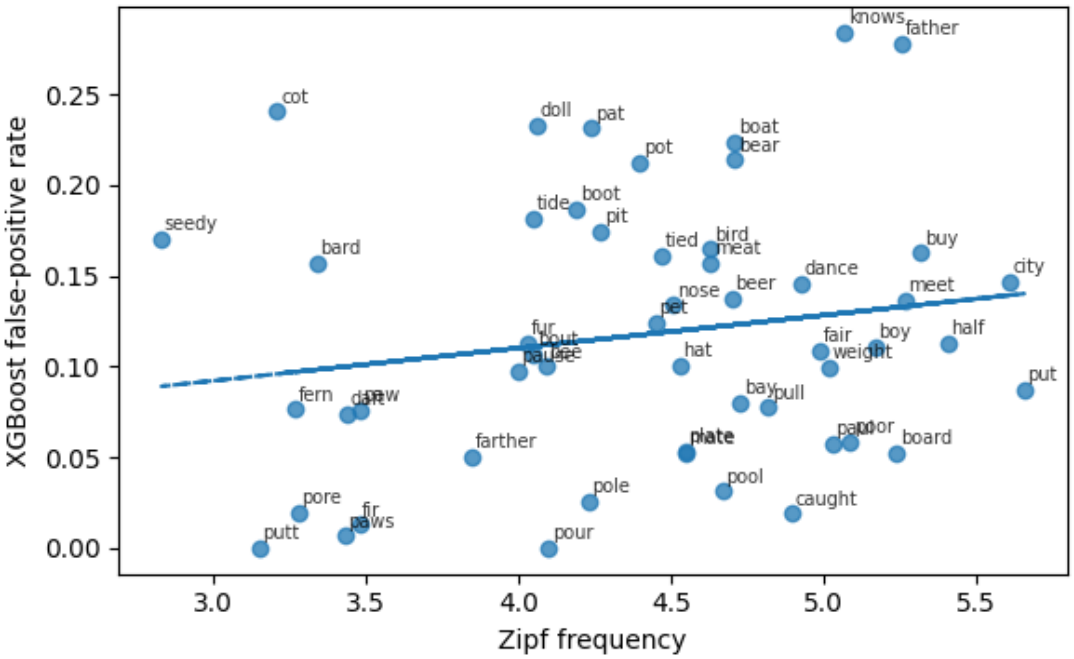


Figure C6: XGBoost FP rate by lexical frequency

Appendix D: Data Statement

Dataset name: MOOC English Pronunciation in a Global World Dataset
Dataset developer(s): Laura Rupp, Luis Morgado da Costa
Data statement author: Ella Heinävaara

Dataset components

The data used in this study consists of three components. Data A consists of learner speech recordings and binary intelligibility annotations. This speaker data was divided into a development set (DEV) and a test set (TEST). Data B consists of manually created GOLD IPA annotations for the 52-word list, based on intelligibility-oriented pronunciation guidelines, particularly Jenkins (2000, 2002). Data C consists of phonological feature-type annotations. This component is still under development and is mentioned here for completeness, but it was not the main object of analysis in the present study.

Curation rationale

The recordings originate from the MOOC *English Pronunciation in a Global World*. The course aims to help learners improve the intelligibility of their English pronunciation. As part of a pronunciation exercise, participants recorded a fixed list of 52 English words designed to represent a range of English phonological contrasts. The recordings were originally collected for educational purposes rather than for the development of automatic speech-processing systems. Their use in the present study therefore constitutes a secondary research application.

Language variety/varieties

The dataset represents highly heterogeneous global English pronunciation. Participants registered for the course from approximately 190 countries. However, country of registration does not necessarily correspond to nationality, first language, or linguistic background. Detailed information regarding speakers' native languages, regional language varieties, and English proficiency levels is not available. Consequently, the dataset should be regarded as representing a diverse sample of L2 English speech rather than any specific language variety.

Speaker demographic characteristics

Detailed speaker demographic information is unavailable. Information regarding age, gender, ethnicity, socioeconomic status, first language, and English proficiency level was not collected or is not publicly available. Consequently, the effects of demographic variables on pronunciation patterns cannot be examined directly.

Participants were learners enrolled in the MOOC *English Pronunciation in a Global World*. Because participation in the pronunciation exercises formed part of the course activities, recordings were produced by students who chose to participate in the course. However, not all registered learners necessarily completed all exercises. The exact number of individual speakers represented in the dataset is currently unknown.

Information regarding the presence of speech disorders is unavailable.

Annotator demographic characteristics

Binary intelligibility labels were assigned by 3 annotators. Most annotations were produced by student research assistants with at least introductory training in linguistics, while a subset of the data was reviewed by an expert annotator

The GOLD IPA transcriptions used in the present study were created separately as part of an internship project by a linguistics student. The transcriptions were developed according to existing pronunciation and phonetic transcription guidelines based on Jenkins (2000, 2002).

Information regarding annotation manuals, annotator training procedures, and inter-annotator agreement measures is currently unavailable. An annotation manual is under development.

Speech situation

The dataset consists of spoken recordings of isolated English words collected in an online learning environment. The recordings are examples of read speech rather than spontaneous speech. Participants recorded themselves asynchronously without the presence of a teacher or researcher. Recording sessions were unsupervised and took place in participants' own environments.

Participants had the possibility to rerecord words and choose which recording to submit. Consequently, the recordings represent self-selected productions rather than necessarily first attempts.

The intended audience of the recordings was primarily other course participants and instructors within the educational setting.

The recordings were collected within the last five years.

Text characteristics

The speech material consists of a fixed list of 52 English words selected to represent a variety of phonological contrasts relevant to intelligibility. The exact source and rationale underlying the selection of the word list are currently unknown.

Because the dataset consists exclusively of isolated lexical items, it contains limited lexical, syntactic, and discourse variation. It does not contain sentence-level or conversational speech.

Recording quality

Recordings were collected remotely through an online course platform. Information regarding microphones, recording devices, and recording environments is unavailable. Recording quality is therefore expected to vary considerably across speakers and may include differences in microphone quality, background noise, and recording conditions.

Collection method

Recordings were obtained through pronunciation exercises integrated into the MOOC *English Pronunciation in a Global World*. Learners completed the recordings independently in unsupervised conditions. Participation in the pronunciation exercises formed part of the course activities, although not all registered learners necessarily completed all exercises.

Participants were able to rerecord words and submit their preferred version rather than being restricted to a single attempt.

Intended use

The recordings were originally collected for pedagogical purposes as part of a pronunciation course focused on intelligibility. The course includes several pronunciation activities, including word-reading tasks and passage reading. The dataset was not originally designed for speech technology research or the development of automatic pronunciation assessment systems. The present study represents a secondary use of the recordings.

Other

For the present study, the original recordings were supplemented with manually curated GOLD IPA transcriptions and binary intelligibility labels.

Three derived resources were used:

- **Development set (DEV)**: used for model development and analysis. This subset contains 2001 recordings, of which 873 were labelled unintelligible and 1128 were labelled intelligible.
- **Test set (TEST)**: used for final evaluation. This subset contains 8215 recordings, of which 3597 were labelled unintelligible and 4618 were labelled intelligible.
- **GOLD IPA lexicon**: a manually curated list of canonical and acceptable pronunciation variants for the 52 target words.

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